LOGO - [Face & Fingerprint Biometric]

[Face & Fingerprint Biometric]

Version V1.25

[Communication Protocol]

Catalogue

Modify log

Date	
January 23rd, 2019	Add and set the type of machine in and out, convenient to display different words Add the option to display the content after successful identification (display: name, Department, job, gender, card number, photo) Add the function of deleting people's photos and fingerprints Increase output type and byte order in Wiegand output proto Modification and reading protocol for increasing threshold of face comparison
May 18th, 2019	Correct the data length and actual mismatch contents in document
May 23rd, 2019	Modify card structure and record type
June 11th, 2019	Add OEM information: manufacturer, website, date of manufacture Modify update record instruction, add predefined instruction parameter error 0x210500
July 15th, 2019	Modify the holiday description in the personnel storage structure and add a holiday corresponding table
July 29th, 2019 V1.9	Delete device validity reading command（0x010700 and 0x010701） Delete authority authentication method（0x030500 and 0x030501） Delete timing lock function（0x030700 and 0x030701） The function definition of access control relay is modified to support bistability or not, 1 is supported, 0 is not supported, and the default is 0（0x030200 and 0x030201）
August 6th, 2019 V1.10	Add the predefined instruction 0x210600 to inform the upper software that encrypted communication is needed
August 23rd, 2019	Add alarm clock settings and master-slave type settings
September 26, 2019	Add automatic test section
October 10th, 2019	The file type of the instruction for file upload is increased by 4-dynamic facial feature code and 3-infrared facial feature code, Return value increased by 2 -- feature code format incorrect, 3 -- personnel photo unrecognized
October 17th, 2019	Since the file upload and download defined in the old version only supports 16m, the fourteenth and fifteenth protocols are added for the large file upload and download of dynamic face machine
November 13rd, 2019	1、 Add the testing scheme for the touch screen of the face machine, and the detailed testing scheme steps;；2、 Modify the authentication record format, delete the reader number and change it to the in out type.
November 22th, 2019	Face machine parameters increase the living detection and distance of recognition
January 2ns, 2020	Add the work type setting of the face machine's fill light, and add the third party HTTP protocol docking function parameter setting
February 10th, 2020	Delete face machine fill light protocol； Increase the machine language and volume, increase the temperature record type.
April 1st, 2020	Add abnormal temperature record type, code: 30
Jan 13th, 2020	V1.15 Add helmet control protocol
Nov 19th, 2020	Add IC card working parameters and visitor password root password parameters

Communication Format Definition

Ver1.15

Baud rate

19200,8,n,1

Communication mode

The following three modes of communication are supported:
1、COM (including using USB connection) （RS485）
2、UDP
The machine does not support TCP connection, only UDP sending and receiving packets.
3、TCP Client Aug 29th, 2020

Format of command issued by PC

Format of communication command issued to the device by the software:

Flag code	Receiver information		Sender information	Control Code			Data Code		Check Code	Flag code
7E	Device SN	Password	Information code	Classification	Command	Parameter	Data length	Data	Inspection value	7E
	16Byte	4Byte	4byte	1Byte	1Byte	1Byte	4Byte	Variable length	1Byte	

Receiver information:

Receive the device information of this communication command, including device Sn and communication password.

Sender information:

Information code: used to identify the serial number of the command. This serial number corresponds to the information code in the return value instruction, which means that the command returns correctly, so as to prevent the mismatch between the return value of the command and the content of the command. For example, to send instruction A, before the controller responds, to send instruction B, the instruction of controller A fails to receive, and instruction B returns successfully. At this time, you can determine whether the instruction returned by the controller is the instruction A sent or the instruction B sent according to the information code in the return value. （It's actually a random number or a serial number）

Special information code

1、0xFFFFFFFF is a broadcast message, which is used when the device sends a real-time message to the upper computer (software).

2、0xBFBFAABB is a UDP broadcast message. The instruction using this message needs to be sent by UDP broadcast. If the device receiving this code needs to reply, it also needs to be sent by broadcast. The target IP of broadcast is 255.255.255.255

2、0xC1ADCCAA is a special code, which is used to obtain the super special password when the RC4 password is forgotten.

Control code:

The specific meaning code of communication command includes classification, command and parameter.

Data code:

Used when a communication command contains data, the data length specifies the number of bytes in the command that contain the data content.

Inspection code:

Except for the flag code and the check code, all bytes in the command want to add and then take the tail sub section.

Data length: data length is determined according to different command specifications, some commands do not have a specified length. The maximum data length of commands without a specified data length cannot exceed 340 bytes.

Example (read controller SN):
7E303030303030303030303030303030FFFFFFFF19883D9001020000000006D7E
Split explanation:
7E--Command header
303030303030303030303030303030 --ascii code of SN (not important when reading Sn, other instructions need to be input correctly)
FFFFFFFF -- Communication password default FFFFFFFF
19883D90 -- Information code (It's actually a random number or a serial number)
010200 -- Control code of corresponding instruction (Refer to the following command table)
00000000--Data length, this is all 0 on no data package part.
6D -- Check sum 303030303030303030303030303030FFFFFFFF19883D9001020000000000 Sum check Take low byte
7E -- Command tail

Format of command issued by Device

Communication command sent by device to software:

Flag Code	Receiver information	Sender information		Control Code			Data Code		Check Code	Flag Code
7E	Information Code	Device SN	Password	Classification	Command	Parameter	Data length	Data	Inspection value	7E
	4byte	16Byte	4Byte	1Byte	1Byte	1Byte	4Byte	Variable length	1Byte	

Receiver code:

See the introduction of sender information code above

Sender information:

Send the device information of this communication command, including device SN and communication password.

Control code:

The specific meaning code of communication command includes classification, command and parameter.

Data code:

Used when a communication command contains data, the data length specifies the number of bytes in the command that contain the data content.

Inspection code:

Except for the flag code and the check code, all bytes in the command want to add and then take the tail sub section.

Information code

The code of the instruction to be responded to (the same as the received instruction code). Event code is generally fixed bit 0xFFFFFFFF.

For example (Read controller SN response):

7E1906677946432D383932304132353035303238 35FFFFFFFF3102000000001046432D383932304132353035303238 35087E

Split explanation:

7E--Command header

19066779 --Network flag issued by software, Software will return whatever it sends.

46432D38393230413235303530323835 - Sn ASCII code of the controller itself

FFFFFFFF -- Communication password

310200 -- Command response code (refer to the command table)

00000010 -- Length of data package part

46432D38393230413235303530323835 --- the content of the data package part (here is the SN instruction, so the data package part is the ASCII code of Sn)

08 -- Sum check

7E -- Command tail

Transcoding

Because 0x7E is used as the start and end flag in the command, the byte 7e cannot appear in the command content but in the command header and end.

The transcoding is as follows:

0x7F 01 = 0x7E

0x7F 02 = 0x7F

Where transcoding is used, before sending the command, check and calculate the content of the command, and then check the data of 0x7e and 0x7F. If 0x7e or 0x7F is found, perform the above formula conversion. After receiving the command, the data should be inverted according to the above formula and then verified and verified.

Predefined commands

Predefined return commands:

Definition of answer OK package:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	01h	00h	00h	No

This definition is directly used in all the subsequent responses and OK responses.

Definition of answer password error:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	02h	00h	00h	No

Definition of answer check error:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	03h	00h	00h	No

Definition of answer IP setting error:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	04h	00h	00h	No

Definition of answer parameter error:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	05h	00h	00h	No

Definition of encryption required for reply communication:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
21h	06h	00h	00h	No

Class I: System Parameters

Preface:：

All categories in this category start with 0x01;；

The beginning of the response in this class is always 0x31；

Generally speaking, the commands and parameters in the reply command need to be the same as those issued.

一、Device SN

Write SN from 1 to 1

That is, the specified Sn must be given, and the device judges the target SN.

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x01	0x00	0x18	0x03C589123E Device SN（0x10） 0x907F78

Explanation： device Sn consists of 16 visible characters (ASC code is between 32-126).

- Reply：OK

Broadcast write SN

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0xC1	0xD1	0xF7	0x18	0xFC656533FF 设备SN（0x10） 0xCF3592

Explanation： device Sn consists of 16 visible characters (ASC code is between 32-126).

The target SN of this command must be sixteen 0x30. This command can only be used in 485.

- Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x02	0x00	0x00	No

Explanation: Reading the SN command does not judge the Sn in the receiver information in the command format

- **Reply：Transmit SN**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x31	0x02	0x00	0x10	SN

二、Communication password

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x03	0x00	0x04	Password

Reply：OK package

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x04	0x00	0x07	0x4643617264597A

Password validation is not required for this command.

- **Answer: communication password**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x04	0x00	0x4	Password

Empty

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x05	0x00	0x07	0x4643617264597A

Password validation is not required for this command. After calling, the password is restored to：0XFFFFFFFF

- Reply：OK

三、IP Parameter

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x06	0x00	0x00	

Answer: network parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x06	0x00	0x89	IP Information

IP information, length 0x89 (137 bytes):

Order	Byte length	Meaning
1	6 byte	MAC address
2	4 byte	IP address
3	4 byte	Subnet mask
4	4 byte	Gateway IP
5	4 byte	DNS
6	4 byte	Backup DNS
7	1 byte	Free
8	2 byte	Free
9	2 byte	Local UDP Monitoring port
10	2 byte	Server port
11	4 byte	Server IP
12	1 byte	Get IP automatically
1	6 byte	MAC address

Send the log monitoring data according to the rules in the current (class 9 data monitoring).

When the IP address is: 0.0.0.0, the IP is restored to the default value: 192.168.1.150

This command can be called by UDP broadcast to read IP parameters, verify Sn and password

DHCP Description:

When the [auto obtain IP] in the network parameter is not equal to 0, the IP address assigned by the DHCP service of the router will be automatically obtained and updated to the [IP address], [subnet mask], [network management IP], [DNS] and other data of the network parameter type

UDP client mode description

The domain name of the server(That is, the first byte is not 0), the domain name resolution shall be carried out immediately, and the resolution result shall be filled in the server IP, and the server IP shall be updated every 2 hours (domain name re resolution)

If only the server IP is filled in, and the server domain name is not filled in, domain name resolution is not required.

When the server domain name or ip is set, the [connection test] messages defined in [9th type of data monitoring] should be sent periodically according to the interval set by [Keep-alive packet sending inter

Write In

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x06	0x01	0x89	IP Information

Reply：OK

五、Get device version number

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x08	0x00	0x00	

Gets the application version number of the device.

Reply：Transmission version number

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x08	0x00	0x0B	Device Version Number

Version number is stored using ASCII characters

Firmware program version number	Firmware program Amendment No	Fingerprint algorithm version number	Version number of face algorithm
Two bytes	Two bytes	Four bytes	Four bytes

六、Get device operation information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x09	0x00	0x00	

Reply：Transmitting information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x09	0x00	0xC	Device information

Device operation information includes，Length 0x12 (18) bytes：

Order	Byte Length	Explain
1	2 Byte	Days of system operation
2	2 Byte	Number of Formatting
3	2 Byte	Watchdog Reset Number
4	6 Byte	Power-on time Time Format：ssmmHHddMMyy，Seconds, minutes, hours, days, months and years Time expressed in BCD code：0x155116160519 -- Express：2019-5-16

		16:51:15

七、Record Storage

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0A	0x01	0x01	0 or 1

0 is full cycle, 1 means full no cycle

- Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0A	0x81	0x00	

- **Reply：Storage Way**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x0A	0x81	0x01	**Storage Way**

八、Online Transaction

Enable

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0B	0x00	0x00	

- Reply：OK

The device remains off by default and is not saved. Refer to Category 10 Messages for the command you receive.

Broadcast Enable

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0B	0x10	0x00	

No SN verification, password verification or response is required when this broadcast is enabled. UDP is acceptable

Close

Control Code	Data Code

Classification	Command	Parameter	Data length	Data
0x01	0x0B	0x01	0x00	

Reply：OK

Broadcast off

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0B	0x11	0x00	

No SN verification, password verification or response is required when this broadcast is enabled. UDP is acceptable

状态 Status

Read real-time monitoring status

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0B	0x02	0x00	

Reply：Monitoring status

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x0B	0x02	0x01	状态

Status: 0 -- monitoring is not turned on; 1 -- monitoring is turned on.

九、Get device status information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0E	0x00	0x00	

Reply：Information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x0E	0x00	6	设备状态

Order	Byte Length	Meaning	Remark
1	1byte	Relay status	0 means COM and NC are normally closed 1means COM and NO are normally closed
2	1byte	Running state	Normally open or normally closed, 0 for normally closed, 1 for normally open
3	1byte	Door sensor switch	0 is off, 1 is on

4	1byte	Door alarm status	(Bit) Mening: Bit0: Illegal authentication alarm Bit1: Door Sensor Alarm Bit2: Force Alarm Bit3: Opening Time Our Alarm Bit4: Blacklist Alarm Bit5: Tamper Alarm Bit6: Fire Alarm
5	1byte	Lock status	0 -- unlocked, 1 -- locked
6	1byte	Monitoring status	0 -- monitoring is not turned on; 1 -- monitoring is turned on.

十、Initialize Device

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x0F	0x00	0x00	

The initialization device includes: clear all personnel, clear all holidays, clear all opening time zones, clear all records, reset the password of management menu, and keep unlocking for 3 seconds.

The opening time zone is 0-24 hours.

- Reply: OK

十一、Search device

UDP search uses broadcast code, and the device should accept broadcast information. Is the packet whose header is the target MAC: 255.255.255.255

UDP broadcast supports【write network parameters】、【read network parameters】

Search for devices with different network identities

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xFE	0x00	0x02	Network identity

This command is a broadcast command， The device receiving this message does not judge whether the target SN of this command matches the local machine, the target password of this command is0xFF...FF。

After receiving this information, the device shall judge whether the network identification of the device is the same as that of the device itself, and only devices with different network identification need to respond.

Search process please refer to：【Appendix V automatic search equipment】

Network ID：a number in the memory of the device. It will be initialized to 0xFFFF each time the device is powered o, the software will generate a group of network IDS immediately each time it searches.

Reply：network parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0xFE	0x00	0x89	IP信息

The response SN is the local SN，and the response password is 0xFF....FF.

Set device network ID

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xFE	0x01	0x02	Network Identity

When this command is received, the device sets the local network identity to the network identity specified in the command.

Reply：OK

十二、Data store operations

Set data store content

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xF0	0x00	0x1E	

Reply：OK

Read data store content

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xF0	0x01	0x00	

Reply：Buffer contents

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0xF0	0x01	0x1E	

十三、Client Keepalive Interval

Set the interval time of keepalive

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xF0	0x02	0x02	Time

Interval time of keeping alive: value 0-65535, unit: second. 0 indicates that the heartbeat keepalive interval parameter is not used.

Reply：OK

Read the interval time of keepalive

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0xF0	0x03	0x00	

- **Reply：Interval time of keepalive**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0xF0	0x03	0x02	Time

十四、Data package encryption switch

Due to business needs, to prevent the data package from being monitored and the communication password from being cracked during transmission.Data package encryption is enabled during data communication. RC4 is used as encryption algorithm. Please refer to the chapter of encrypted communication in the appendix for details of communication format change after encryption. The RC4 key is set independently with 32 bytes (256bit). **Initialization default value：off**

The communication encryption switch is affected by the initialization instruction. The initialization communication encryption switch will be released, that is, the communication is not encrypted.

Set switch:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x17	0x00	33	Function switch (1) + 32 byte key

Reply: OK

Reading parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x17	0x01	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x17	0x01	33	Function switch (1) + 32 byte key

十五、Local Identity

Function Description:

<u>Only valid for access controller, effective when access controller, fingerprint and face biometric as reader</u>

Setting:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1C	0x00	62字节	1 byte of the local door number Local name 60 bytes In out category 1 byte

Reply: OK

Local door number: value range 1-4;

Local Name, which is used to display the name of the machine on the screen;

The encoding of simplified Chinese mode is Unicode

In/Out category: 0--In; 1--Out

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x01	0x1C	0x01	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x1C	0x01	62	1 byte of the local door number Local name 60 bytes In out category 1 byte

十六、Wiegand Output

Function Description:

Only valid for access controller, effective when access controller, fingerprint and face biometric as reader.

Setting:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1C	0x02	0x04	Reading Byte+WG Output+WG byte order +Output data type

Reply: OK

Read card bytes

Byte Value	Meaning
1	WG26(Three bytes)
2	WG34(Four bytes)
3	WG26(Two bytes)（Cancelled）
4	WG66(Eight bytes)
5	Disable

WG output function switch: 1--Enable; 2--Disable

WG byte order: 1--high order in front and low order after; 2-- low order in front and high order after

Output data type: 1 -- output user number; 2 -- output personnel card number

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1C	0x03	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x31	0x1C	0x03	0x04	Reading Byte+WG Output+WG byte order +Output data type

十七、Comparison threshold of face and fingerprint

Function Description:

Only valid for access controller, effective when access controller, fingerprint and face biometric as reader

Setting:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1C	0x04	0x02	Fingerprint comparison threshold 0x1 Threshold value of face comparison 0x1

Reply: OK

Threshold: the value that controls the accuracy of face / fingerprint comparison.

Threshold value range: 100-1.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1C	0x05	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x1C	0x05	0x02	Fingerprint comparison threshold 0x1 Threshold value of face comparison 0x1

十九、Admin Menu Password

Set admin password

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x01	0x1D	0x00	04	Password (4)

Reply: OK

The password is stored in BCD format with a minimum of 4 digits. Maximum 8 digits, less than 8 digits, followed by F

For example, password 1234, value: 0x1234FFFF

Read admin password

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1D	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x1D	0x01	04	Password(4)

二十、OEM information

Set OEM information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1E	0x00	127	

Reply: OK

Serial Number	Type	Length	Format
1	Name of manufacturer	60 Byte	Gb2312
2	Website	60 Byte	Gb2312
3	date of manufacture	7 Byte	MM / DD / YY/ HH/M / S Example 0x20190611123005 June 11, 2019 12:30:05

Read OEM information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1F	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x31	0x1E	0x01	127	OEM Information

二十、Set the master-slave type of the machine

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1F	0x00	1	Master-slave type

Reply：OK

Master-slave type: 1 -- host (default); 2 -- slave

主 Master After the master-slave type is set, it cannot be restored through initialization.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x1F	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x1F	0x01	1	Master-slave type

二十二、Device alarm clock

Set alarm clock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x20	0x00	24*3=72	Alarm list

Reply：OK

There are 24 groups of alarms, with 3 bytes for each group

Alarm structure description：

Order	key	Explain	Bytes
1	hour	In the alarm time H，24-hour system；value range 0-23	1
2	Minute	In the alarm time M；value range 0-59	1
3	Times	In the alarm time，Unit is second；value range 0-60	1

Explain	If the alarm time is 12:00, the times is 5, So the alarm starts at 12:00 at noon and rings for 5seconds The alarm clock output includes the machine itself playing sound and output an alarm clock signal for external devices

Reading alarm clock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x20	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x20	0x01	24*3=72	OEM Information

二十三、Face machine live detection

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x21	0x00	1	Switch

Reply：OK

Livenesss Detection Type：0--Disable；1--Enable；

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x21	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x21	0x01	1	Switch

Setting Face Recognition Threshold

Added in March 25th, 2022

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x21	0x02	1	Threshold

Reply：OK

Face Recognition Threshold： 1-99

Reading Liveness Detection Threshold

Added in March 25th, 2022

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x21	0x03		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x21	0x03	1	Threshold

二十四、Face Recognition Distance

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x22	0x00	1	Distance

Reply：OK

Distance：1--Short distance（0.2-0.5m）；2--Middle distance（0.2-1.5m）；3--Far Distance（0.2-1.5m above）

Default: middle distance

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x22	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x22	0x01	1	Distance

二十五、HTTP protocol docking parameters of the third-party platform of face Terminal

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x23	0x00	260	

Reply：OK

Docking Parameters

Byte	Explain	Value Range
1	Switch	1--Enable 0--Disable（Default）
1	Platform Type	1--cloud building network
1	URL length	1-255
255	URL	String to store the web address Ascii code
2	Parameter length	Platform parameter length

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x23	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x23	0x01	260	Parameter

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x23	0x02	N 1500 bytes maximum	json string of docking parameter

Reply：OK

Json string of cloud building network platform

{“appId”:” f803b341ca504dfe9d019c323e407697”,

“secretKey”:” 00b766006b474837afb25a3ffd79331d”}

Reading

Control Code	Data Code

Classification	Command	Parameter	Data length	Data
0x01	0x23	0x03		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x23	0x03	N	Json string of docking

二十六、Setting Language

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x25	0x00	1	Type

Reply：OK

1 - Chinese
2 - English
3 - Traditional Chinese
4 - French
5 - Russian
6 - Brazil
7 - Spanish
8 - Italian
9 - Japanese
10 - Korean
11 - Thai
12 - Arabic
13 - Portuguese
14 - Turkish
15 - Indonesia
16 - Ukrainian

Default Value： 1

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x25	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x25	0x01	1	Type

二十七、Set Device Volume

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x26	0x00	1	Volume

Reply：OK

Volume value range：0-10；0--turn off sound；10-- Maximum sound

Default Value：10

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x26	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x26	0x01	1	Volume

二十八、Set the supplementary light mode of the face terminal

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x27	0x00	1	Mode

Reply：OK

Value Range of Mode：1--Always on；2--Open when personnel are detected；0--Always off

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x27	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x27	0x01	1	Mode

二十九、Setting Mask Identification

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x28	0x00	1	Switch

Reply：OK

Value：0--Disable；1--Enable

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x28	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x28	0x01	1	Switch

三十、Setting the temperature detection and format of the face terminal

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x29	0x00	2	Temperature detection switch+display format

Reply：OK

Temperature Detection Switch：0--Disable；1--Enable

Display Format：1--Degrees Celsius (default)；2--Fahrenheit degree；

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x29	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x29	0x01	2	Temperature detection switch+display format

三十一、Setting the Weather Information of the Face Terminal

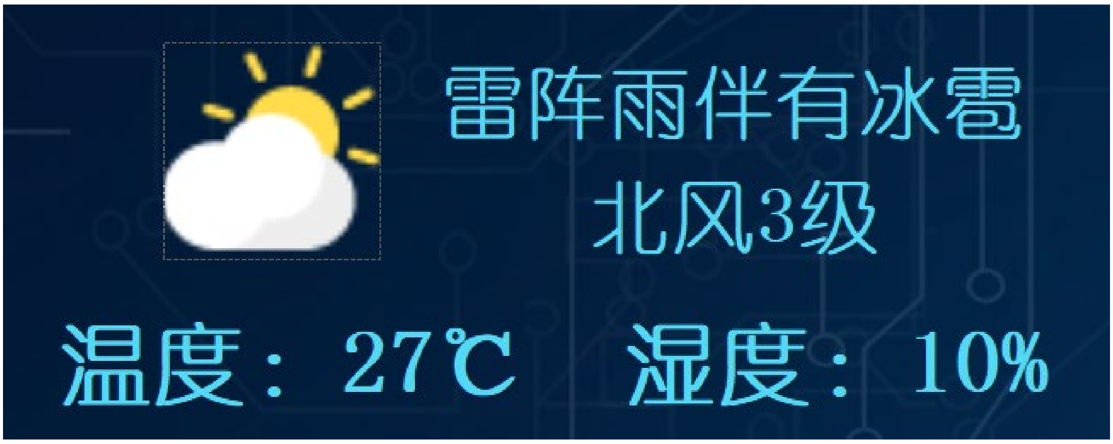
Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2A	0x00	114	Switch+weather information (113)

Reply：OK

Weather Switch：0--Disable；1--Enable

Temperature	Humidity	Weather Status	Wind Direction	Wind-force	Icon	Release time
2 bytes	2 bytes	6 0bytes	40 bytes	1 byte	1 byte	7 bytes
Unit: ℃ 10 times of the actual value 27.5 ℃ stored as 275	10 times of the actual value 85.6% stored as 856	Unicode Example: Thunderstorm accompanied by hail	Unicode Example: North wind	0-20	0-46 See appendix, weather icon corresponding table	yyyyMMddHHmmss BCD



Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
Control Code			Data Code	

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2A	0x01	114	Switch+weather info(113)

三十二、Setting the Cleaning Indicator Switch

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2B	0x00	1	开关

Reply：OK

Switch:：0-turn off the light (default value);；1-Light on

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2B	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2B	0x01	1	Switch

三十三、Setting Do Not Disturb Light Switch

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2C	0x00	1	Switch

Reply：OK

Switch：0-Light off（default value）；1-Light on

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2C	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2C	0x01	1	Switch

三十四、Setting the Temperature Alarm Threshold

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2D	0x00	2	Alarm Threshold

Reply：OK

Threshold storage format; Real temperature *10; For example, 37.5 is stored as 375;

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2D	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2D	0x01	2	Threshold

三十五、Temperature Detection Information Display Switch

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2E	0x00	1	1 or 0

Reply：OK

0--Do not display the temperature； 1--Display the temperature

When the display of body temperature is prohibited, the body temperature record shall still be recorded after legal verification.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2E	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2E	0x01	1	Switch

三十六、SMS after Legal Verification

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2F	0x00	60	Unicode字符串

Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x2F	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x2F	0x01	60	Unicode字符串

三十七 Client Network Parameters

一、Modify Network Server Parameters

The protocol is used to modify the network server parameters in the network IP parameters to avoid modifying the IP address

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x00	105	Server Parameters

Reply：OK

Order	**Byte length**	**Meaning**
1	**2 bytes**	**Server Port**
2	**4 bytes**	**Server IP**
3	**99 bytes**	**Server Domain**

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x30	0x01	105	Server Parameters

二、Client Communication Mode

When the protocol is used to modify the client mode, the server connection ways are TCP, UDP, and Disable

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x02	1	Client Mode

Reply：OK

Value	Meaning
0	Disable
1	UDP
2	TCP Client

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x03		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x30	0x03	1	Client Mode

三、Send the keepalive package immediately

Device send the keepalive package immediately

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x04	0	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x30	0x04	1	Status

Status Description:

Value	Illustration
1	Sending Succeeded
2	Server Parameter not set
3	Server Parameter error
4	Server Connection failed（TCP）
5	No response from the server
6	Network parameter setting error
7	Network cable is not connected
8	Wifi Not connected

四、Get Server Connection Status

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x05	0	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x30	0x05	13	

Return Content:

Order	Byte Length	Meaning
1	1 Byte	Client Mode 0-Disable 1-UDP 2-TCP Client 3-TCP Client + TLS 4-MQTT（TCP Client） 5-MQTT（TCP Client） + TLS
2	4 Bytes	Server IP -- IP after domain name resolution
3	7 Bytes	The sending time of the last keepalive package BCD：yyyyMMddHHmmss

4	**1 Byte**	**Connection Status** **0--TCP Client Not connected;** **1--TCP Client Connected;** **2--UDP Client No connection status** **255--Disabled**

五、Reconnect the server now

Let the device reconnect to the server immediately. When it is connected, please disconnect it immediately and reconnect it; It is applicable to TCP Client mode. In UDP Client mode, only keepalive package are re-transmitted.

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x30	0x06	0	

Return Value

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x30	0x06	1	

Status Description:

Value	**Illustration**
1	**Reconnected** **UDP indicates that the keepalive package has been sent**
2	**Server Parameter not set**
3	**Server Parameter error**
4	**Server Connection failed（TCP）**
5	**No response from the server**
6	**Network parameter setting error**
7	**Network cable is not connected**
8	**Wifi Not connected**

三十八 Reboot device

Restart the face terminal

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x31	0x00	0	

Reply：OK

三十九 Get whether support the 64 floors of elevator board

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x32	0x00		

应答：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x32	0x00	1	1--Support 0--Not Support

四十 Authentication Mode

Supported by FC8300 version 5.36 or above on February 22, 2021

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x33	0x00	1	Authentication Mode

Reply：OK

Authentication Mode Description

Value	Meaning
1	Standard Mode Default Value Supported authentication media: [Swipe Card] or [Face] or [User ID+Password]
2	Face+password After face recognition, you need to enter your personal password. If do not have personal password, can passed directly Supported authentication media: [Swipe Card] or [Face+Password] or [User ID+Password]
3	Card+face The face must be verified after swiping the card, and the card can be swiped directly if there is no person's photo registered on the card. If only registered photo, the card is not registered, It will not recognize faces

	[User ID+Password] verification is not supported Supported verification media: [Swipe Card] or [Swipe Card+Face]
4	Multi-person attendance Multiple personnel can be detected at the same time. Temperature measurement is not supported Supported authentication media: [Swipe Card] or [Face] or [User ID+Password]
5	Comparison between people and identification card In this mode, only ID card swiping is supported before face verification In this case, it is not necessary to compare the face database, only the photos read from the ID card This mode does not support swiping, face verification, user ID+password Authentication comparison only generates anonymous records (the user number is 0, but the type is [Authentication Comparison])

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x33	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x33	0x01	1	Authentication Mode

四十一 Save Photo Switch

Supported by FC8300 version 5.36 or above on February 22, 2021

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x34	0x00	1	0 or 1

Reply: OK

Switch Description

Value	Meaning
1	Enable Default Value Save live photos during face recognition
0	Disable During face recognition, only records are saved, not on-site photos

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x34	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x34	0x01	1	Switch

四十二、QR Code Generation Switch for Recognition Result Query

Supported by FC8300 version 5.36 or above on February 22, 2021

After the function is enabled, face recognition or card swiping for taking photos will generate a QR code for mobile phone scanning and query

The QR code contains this identification information (record serial number, user number, name, time, device name, in/out, temperature, event type, device SN)

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x35	0x00	Switch 1 + Server URL 120	Switch：0 or 1 Server URL：AscII Code

Reply：OK

Switch Description

Value	Meaning
1	Enable After recognition,the identification result query code will be displayed on the upper right corner of the screen
0	Disable Default Value

Server URL:

Format Example：http://yun.pc15.net/FaceRecord/QueryQRcode.html

The actually generated QR code content is URL+parameter (parameter name in lower case)

Parameter Definition:

Parameter	Meaning	Data type

serialnum	Record number	Value
usercode	User ID	Value
name	User Name	Utf8 encoded characters
time	Time Format is： yyyy-mm-dd HH:mm:ss	utf8 encoded characters
drivename	Equipment Name (Equipment Title)	utf8 encoded characters
checktype	Type of Access	Value 1 -- Means entering the door; 2 -- Indicates going out
bodytmp	Temperature The value is the actual temperature * 10	Value Unit Celsius degree
eventcode	Event Code Please check the format definition of authentication record	Value
drivesn	Device sn	utf8 encoded characters

Example of actual content of QR code

http://yun.pc15.net/FaceRecord/QueryQRcode.html?serialnum=1&usercode=1&name= 测 试 &time=2021-02-22 15:38:01&drivename=人脸机&checktype=1&bodytmp=365&eventcode=3&drivesn=FC-8300T20210205

URL Content after URL encoding

http://yun.pc15.net/FaceRecord/QueryQRcode.html?serialnum=1&usercode=1&name=%E6%B5%8B%E8%AF%95&time=2021-02-22%2015:38:01&drivename=%E4%BA%BA%E8%84%B8%E6%9C%BA&checktype=1&bodytmp=365&eventcode=3&drivesn=FC-8300T20210205

Generated QR code

Reading

Control Code	Data Code

Classification	Command	Parameter	Data length	Data
0x01	0x35	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x35	0x01	1+120	Switch+URL

四十三、Photosensitive Mode

Supported by FC8300 version 5.36 or above on February 22, 2021

The brightness of the picture captured by the camera can be adjusted

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x36	0x00	1	1 or 2

Reply：OK

Switch description

Value	Meaning
1	Standard Default
2	Brighten
3	Dark

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x36	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x36	0x01	1	Mode

四十四、Offline Message Pushing

September 29th, 2021

After the switch is turned on, offline messages will be pushed preferentially while networking

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x37	0x00	1	0 or 1

Reply：OK
Switch Description
Default value is 0, and offline messages are not pushed

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x37	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x37	0x01	1	Switch

四十五、Offline Card Authorization

December 23rd, 2021
After the switch is turned on, need to set the parameters of the built-in card reading module

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x38	0x00	1+4+4	1-byte switch 4-byte device number 4-byte group number

Reply：OK
Switch description
After initialization, the default value is 0, and the offline card function is not enabled.
Set parameters for the card reading module and turn off the offline card reading function.
After the switch is turned on, set the parameters of the built-in card reading module, and set them once after each boot.
功能开启后，内置读卡模块读取到卡号后会传输给人脸机，人脸机收到卡号后，直接开门，
如果卡片有登记就记录，卡片未登记则记录为免验证开门。
同时上传的实时监控记录需要修改格式。（0x190101）

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x38	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x38	0x01	9	Switch

四十六、Helmet detection

January 12th, 2022

After the switch is turned on, it is necessary to detect whether the personnel identification is wearing a helmet

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x39	0x00	1	1 Byte Switch

Reply：OK

Switch description

After initialization, the default value is 0 and is not enabled.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x39	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x39	0x01	1	Switch

四十七、Cloud building platform protocol module

March 25th, 2022

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3A	0x00	1	1 Byte Switch

Reply：OK
Switch description
After initialization, the default value is 0 and is not enabled.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3A	0x01		Switch

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x3A	0x01	1	

Retrieving Personnel

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3A	0x02		

Reply: OK

四十八、4G Network Module

July 20th, 2022

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3B	0x00	1	1 Byte Switch

Reply：OK
Switch description
1--Enable 4G network module；0--Disable the 4G network module.

After setting the parameters, the device will restart.And after the 4G network module is enabled, the device will load the driver of the 4G network module and interoperate with the 4G network module, which will increase the device boot time (the 4G network module will increase the boot time by more than 16 seconds)

After the 4G network module is disabled, the 4G network module will not be loaded when the device is powered on, and will not interoperate with the 4G module when the device is running.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x01	0x3B	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x3B	0x01	1	Switch

四十九、Third Party Platform Docking Parameters

September 9th, 2022

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3C	0x00	N	Uft-8 String

Reply：OK

Key adopts hump style, and the first letter of each word is capitalized

Parameter utf-8 encoded string format: key value pair format, separated by English colon and space, and each line of a key value pair ends with\r\n（0x0D0A）

Refer to HTTP Header Model

Key: Value \r\n

Key: Value

When sending the command, only the corresponding platform parameters are set specifically, and non platform parameters are not transferred. For example, the third-party platform is disabled when sending the command

platformtype:0

Example: Send to open Fuyang platform

platformtype: 2

fy_ipaddr: 192.168.1.55

fy_aeskey: 123123123123123123123

fy_udp_port: 1111

fy_tcp_port: 2222

fy_ankang: 0

Parameter Dictionary

Key	Value Type	Illustration	Real Name System Platform
PlatformType	Number	0-Disable third-party platforms	Platform Type
		1-Cloud Building Platform	
		2-Real name system in Fuyang, Anhui Province	

		3-Hunan real name system	
fy_ankang_cityno	character string	City code	Fuyang real name system Ankang Code
fy_ankang_client_key	character string	Manufacturer Key	
fy_ankang_client_id	character string	Manufacturer ID	
fy_ankang_url	character string	Ankang Code Platform URL	
fy_ipaddr	character string	Fuyang server IP	Fuyang workers' real name system
fy_aeskey	character string	Device AES Secret Key	
fy_udp_port	Number	Fuyang server UDP port number	
fy_tcp_port	Number	Fuyang server TCP port number	
fy_ankang	Number	Open Ankang Code 0 or 1	
hn_deviceToken	character string	Hunan Device token	Hunan workers' real name system
hn_applyId	character string	Supplier token	
hn_supplierCode	character string	Supplier code	
hn_url	character string	Hunan server address	
hn_deviceID	character string	Device ID	

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3C	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x3C	0x01	N	Uft-8 String

五十、Root Password of Visitor Password

November 17th, 2022

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3D	0x00	4	BCD

Reply：OK

The visitor's password is 8 numbers, and the BCD code format 0x12345678 is used during transmission, indicating that the visitor's password is 12345678

If the number is less than 8, fill in F, and the minimum number is 4. That is, it can be 0x1234FFFF, so the actual effective password is 1234

If it is 0xFFFFFFFF, it means that the guest password function is disabled

The record of the visitor's password after opening the door is in the type of punch in record, and the record code is 31

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3D	0x01		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x3D	0x01	4	BCD

五十一、Mifare card working parameters

November 19th, 2022

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3E	0x00	39 bytes	User card parameters+setting card parameters

Reply: OK

Offline door opening card is divided into two types

1、User Card

(1) The user card has only one sector to store the user card data. For the card data structure, refer to [Appendix 11 IC Card Data Structure - User Card]

(2) User Card Controlled Card Function Restrictions

(3) The sector password of the user card can be a dynamic password

(4) User card can be limited to CPU compatible card

2、System parameter setting card

(1) The parameter setting card is divided into three categories

① Access control parameter setting card

② Card Number Permission Setting Card

③ Opening time zone setting card

(2) When using the parameter setting card, you need to press the password key to enter the function menu and then read the card to correctly read the parameter setting card

(3) Reference document for data structure in parameter setting card [Appendix 12 IC card data structure parameter setting card]

User card parameter list 30 bytes

Bytes	Meaning
1	IC card sector code

6	IC card sector password The password in the final card is affected by the [Dynamic Password] switch
1	IC card sector password type: 1-A Encrypted 2-B Encrypted
2	Local first level identification code Area code Value range of each identification code 1-72
2	The machine's secondary identification code Building code The value range of each identification code is 1-72
2	The three-level identification code of this machine Floor code The value range of each identification code is 1-72
2	The machine's four-level identification code door number The value range of each identification code is 1-72
1	Forced to read IC card sector data 1--Compulsory reading, the IC card must be decrypted with the specified IC card sector password, and then read the sector data. If the sector password identification fails, ignore this card reading and do not record the event; 0--Do not read data compulsorily, and when the IC cannot be decrypted with the specified sector password, only the serial number of the IC card will be read, and the locally registered card will be detected and matched.
1	IC card card number sequence. 0--take the last three bytes or two bytes; Example: card number 0x659197D2; two bytes: 0x97D2; three bytes: 0x0x9197D2 1--Take the first three or two bytes; Example: card number 0x659197D2; two bytes: 0x6591; three bytes: 0x0x659197
1	Only support CPU compatible card switch 1-on, 0--disable After this switch is turned on, the use of standard MF1 cards will be prohibited, and only CPU-compatible MF1 cards are supported.
1	Use dynamic password 1-open, 0-disable After enabling, the password of the user card sector is a dynamic password. For the algorithm of the dynamic password, refer to [Appendix 10 IC Card Sector Dynamic Password]
1	Enable control card function 1-open, 0-disable After enabling the card control function, you need to verify the sector where the card control function is located when reading the card, and verify whether the first 8 bytes of the sector where the card control function is located is 0x88992B20C80D825D After the verification is passed, it is allowed to continue to check the parameters such as door opening authority, otherwise the device ignores this card reading
1	Control card function--IC card sector number
6	Card control function--IC card sector password The final password in the card is affected by the [Dynamic Password] switch
1	Control card function--IC card sector password type:

	1-A encryption 2-B encryption
1	Rolling code function 1-on, 0--disable This function can only be enabled after the control card function is enabled The operating logic of the rolling code refers to [Appendix 13 IC Card Rolling Code Function]

System setting card parameters 9 bytes

Byte	Implication
1	Set the starting sector number of the card
6	Sector password
1	IC card sector password type: 1-A encryption 2-B encryption
1	After the configuration is complete, clear the setting card 1-open, 0-disable After reading the content of the setting card, clear the card data to avoid re-importing with the same configuration next time.

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x01	0x3E	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x31	0x3E	0x01	39	Byte

The second type of time and date

Foreword:

The categories in this category all start with 0x02;

The beginning of the response in this class is always 0x32;

Generally speaking, the command and parameters in the response command need to be the same as the issued command.

1. Read device time

Control Code			Data Code	
Classification	Order	Parameter	Data length	Data
0x02	0x01	0x00	0x00	

Response: time

Control Code			Data Code	
Classification	Order	Parameter	Data length	Data
0x32	0x01	0x00	0x07	Time

Time format: ssmmHHddMMWWyy, second minute hour day month year

Day of the week: 1 means Monday; 2 means Tuesday. . . . 6 means Saturday; 7 means Sunday;

2. Write device time

Control Code			Data Code	
Classification	Order	Parameter	Data length	Data
0x02	0x02	0x00	0x07	Time

Response: OK

Time format: ssmmHHddMMWWyy, second minute hour day month year

School Time broadcast

Control Code			Data Code	
Classification	Order	Parameter	Data length	Data
0x02	0x02	0x01	0x07	Time

There is no need to reply to this command, no need to verify SN, no need to verify password, no need to reply

Class III Access Control Parameters

Preface:

All categories in this category start with 0x03;

The beginning of the response in this class is 0x33;

Generally speaking, the commands and parameters in the reply command need to be the same as those issued.

Generally speaking, 0 means invalid, 1 means valid

一、Card number byte

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x01	0x00	0x00	

Reply: Card byte parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x01	0x00	1	Parameter

card number bytes:

Byte Value	Meaning
1	WG26(Three Bytes)
2	WG34(Four Bytes)
3	WG26(Two Bytes)（Cancelled）
4	Disable
5	8Byte

Write card number byte parameter

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x01	0x01	0x01	Parameter

Reply: OK

二、Door relay(bistable)

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x02	0x00	0x00	

- **Reply：Relay bistable function switch**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x02	0x00	0x01	Parameter

Whether the relay supports bistability, 1 is supported, 0 is not supported, and the default value is 0

Write In

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x02	0x01	0x01	Parameter

- Reply：OK

三、Remote Unlock

1、Unlock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x03	0x00		

- 应答：OK

2、Open door_with verification code

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x03	0x80	0x01	Verification Code

- Reply：OK

Verification Code: reset to FF when the hardware is powered on, value: 1-254, function, verify whether the command is sent repeatedly.

四、Close Door

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x03	0x01		

Reply：OK

When the door is closed, if the door is set to normally open, the normally open state will be ended.

五、Setting door normally opened

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x03	0x02		

Reply：OK

Power off does not save state.

六、Locked

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x04	0x00		

Reply：OK

Do not accept the operation of card reading, fingerprint, face, password and door opening button after locking

七、Unlock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x04	0x01		

Reply：OK

九、Door normally opened parameters

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x06	0x00		

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x06	0x00	0xE2	Port No.+Parameters

Data information:

1byte	1byte	224byte
Enable	Normally open trigger mode	Normally open time zone 7*8*4

Normally opened trigger mode:

Value	Explain
1	After the legal authentication is passed, it can be opened normally in a specified period of time
2	If the authorization is marked as the normally open privilege, it can be normally open after being authenticated within a specified period of time
3	Automatic switch, automatically open and close the door at the time.

Normally Opening Time zone:

It is allowed to enter the normally open time zone. After the time overed, the door will be locked automatically according to the setting.

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x06	0x01	0xE2	Parameters

Reply: OK

十一、Unlock Output Time

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x08	0x00		

Reply：Output time

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x08	0x00	0x2	Time

Port number 1 byte, output time 2 bytes, maximum 65535 seconds. 0 for 0.5 seconds

Setting the output time when unlocking

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x08	0x01	0x02	时长

Reply：OK

十二、Open without verification

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x11	0x00	1+1+1	Open without verification enabled+ Enable automatic registration+ Automatically register period number

Reply：OK

Reading open without verification

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x11	0x01		

Reply：open without verification parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x11	0x01	0x3	Parameter

十三、Voice Broadcast Function

Setting Voice Broadcast

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x13	0x00	1	Enable(1)

1 for on, 0 for off

Reply：OK

Reading voice broadcast

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x13	0x01		

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x13	0x01	1	Enable(1)

1 for on, 0 for off

十四、Repeat validation permission interval

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x14	0x00	0x04	Interval Parameters

Data format：

1Byte	1Byte	2Bytes
Enable Switch 0(Close) or 1(Open)	Detection Mode	Interval Time

Detection Mode：

Value	Explain
1	Record the event, not open door, have prompt
2	Do not record the event, not open door, have prompt
3	No response, no prompt

The interval is 2 bytes, with a maximum of 65535 seconds. 0 means unlimited.

- Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x14	0x01	0x00	

- **Reply：Repeat validation parameters**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x14	0x01	0x04	Interval Time

十五、Prompt for permission expiration

When register cards after being enabled, if the validity period is less than one month, it will be prompted that the validity period is less than one month

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x15	0x00	0x02	Enable（1） Validity threshold（1）

First byte: whether it is enabled, 0 is not enabled, 1 is enabled.

Second byte: validity threshold

The default value is 30 days after initialization of this threshold value；when the validity period of the user's card is lower than this value, swipe card can open the door, but the record event is changed to the record code【24--Legal opening, the validity period is ready to expire】. And play voice reminders.

24	Pass validation - validity is about to expire

Value range：1-255。0--means close。

Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x15	0x01	0x00	

Reply：Expiration prompt parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x15	0x01	0x01	Enable

十六、Door open verification method

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x16	0x00	0x01	Authentication method (1)

Ways of identifying:

1-face/password/swipe card/fingerprint (default value)

2-Face/swipe card/fingerprint + password (password is required after face verification, password is required after fingerprint verification, and the door is opened directly without a password)

Reply：OK

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x16	0x01	0x00	

Reply：**Expiration prompt parameter**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x16	0x01	0x01	Ways of identifying

Elevator Mode Parameters

Elevator working mode

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x21	0x00		

- **Reply: elevator mode**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x21	0x00	1	0 or 1

0 - disable elevator mode

1--Start elevator mode

Writing

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x21	0x01	1	0 or 1

- Reply：OK

Elevator relay (COM&NO normally closed, COM&NC normally closed)

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x21	0x00		

- **Reply: relay type**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x22	0x00	64	Parameter

Each byte indicates the relay type of a port, the first byte indicates port 1....port 64
Relay types are divided into: 1. COM&NC normally closed (default value); 2. COM&NO normally closed;

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x22	0x01		

- **Reply: Relay status**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x22	0x00	64	Parameter

Each byte indicates the relay status of a port, the first byte indicates port 1....port 64

The status of the relay is divided into: 1. COM&NC normally closed (default value); 2. COM&NO normally closed;

Writing

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x22	0x02	64	Parameter

- Reply：OK

Each byte indicates the relay type of a port, the first byte indicates port 1....port 64

Open the door remotely

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x23	0x00	64	

- Reply：OK

Each byte represents a port, the first byte represents port 1....port 64

A byte value of 0 indicates that this port is ignored, and a byte value of 1 needs to drive this port relay

Close the door

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x23	0x01	64	

- Reply：OK

When closing the door, if the door is set as normally open, it will end the normally open state.
Each byte represents a port, the first byte represents port 1....port 64
A byte value of 0 means ignore this port, a byte value of 1 requires an action

Remote setting door normally open

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x23	0x02	64	

Reply：OK
Power off does not save state.
Each byte represents a port, the first byte represents port 1....port 64
A byte value of 0 means ignore this port, a byte value of 1 requires an action

Remote lock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x24	0x00	64	

- Reply：OK

Card reader, fingerprint, face, password, and door open button operations are not accepted after locking
Each byte represents a port, the first byte represents port 1....port 64
A byte value of 0 means ignore this port, a byte value of 1 requires an action

Remote unlock

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x24	0x01	64	

Reply：OK

Each byte represents a port, the first byte represents port 1....port 64

A byte value of 0 means ignore this port, a byte value of 1 requires an action

Timing and always-on parameters

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x26	0x00	1	The port number

Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x26	0x00	0×E3	Parameter

Data information:

1byte	1byte	1byte	224byte
The port number	Enable	Normally open trigger mode	Normally open time zone 7*8*4

Normally open trigger mode:

Value	Explain
1	After the legal certification is passed, it can be normally opened within a specified period of time
2	Those marked as normally open privileges in the authorization can be normally opened after passing the authentication within a specified period of time
3	Automatic switch, automatically open and close the door when the time comes.

Normally open hours:

It is allowed to enter the normally open period, and the door will be automatically locked according to the setting after the time passes.

Setting

Control Code	Data Code

Classification	Command	Parameter	Data length	Data
0x03	0x26	0x01	0×E3	Parameter

- Reply：OK

Output duration when unlocking

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x28	0x00		

- **Reply: output duration**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x33	0x28	0x00	64*2	Duration

Each group of 2 bytes represents the output duration of a port, the sequence port 1....port 64

Output duration value range: 2 bytes, maximum 65535 seconds. 0 means 0.5 seconds.

Set the output duration when unlocking

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x03	0x28	0x01	64*2	Duration

- Reply：OK

第四类 报警参数

一、Fire Alarm

Fire Alarm Notice

Inform the device to trigger the fire alarm

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x01	0x00	0x00	

- Reply：OK

Function Switch

March 25, 2022 Added

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x01	0x01	0x01	Function switch

- Reply：OK

Function switch: 1--function on, 0-function off

二、Blacklist Alarm

Setting

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x2	0x00	0x01	开(1)或关(0) On(1) or off(0)

- Reply：OK

Indicates whether to enable the blacklist alarm function，if this function is turned off, it indicates that the card swiping in the blacklist will not alarm and only record.

This feature defaults：Open

Reading

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x2	0x01	0x00	

- Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x2	0x01	0x01	Open(1)or close(0)

三、Tamper Alarm Function

Setting Switch

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x3	0	1	Function Switch（1）

Reply：OK

Value Range：0--Close function，1--Open function

After the tamper switch is disconnected, the tamper alarm of the device will be recorded, which will be displayed when getting the status, 当 When the tamper switch is short circuited, it will automatically turn off the alarm，（In addition, do not turn off unless the anti disassembly function is turned off）

Default value：0--close function.

Reading Parameters

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x3	0x01	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x3	0x01	1	

四、Illegal verification alarm

Only for card reading, password, fingerprint. Invalid face does not need to alarm.

Set illegal verification alarm switch

Control Code			Data Code	

Classification	Command	Parameter	Data length	Data
0x04	0x04	0x00	0x02	Switch(0 or 1)+times

Data format

First byte	Second byte
Enable?	Times of illegal verification Value range：0-255, 0--Alarm for reading card one time

Illegal verification reached a certain number of alarm

Default value：0--alarm if fail to verify once.

- Reply：OK

Read illegal verification alarm switch

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x04	0x01		

- **Reply：illegal verification alarm number**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x04	0x01	0x02	

Refer to the writing parameter format

五、Duress alarm password

Set duress alarm password

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x05	0x00	0x06	0 or 1+4 digits password+mode

Data format：

1byte	4byte	1byte
Effective or not?	Password	Alarm mode

Alarm mode：

Value	Explanation

1	Locked, alarm output
2	Unlocked, alarm output
3	Lock the door, alarm, only software unlock

密码：Password: format is BCD code, that is to say, 0x12345678 means password is 12345678, password is 8 digits, the value of each digit is 0-9, if the password is less than 8, write the placeholder 0xF in the following space, for example, password is 8765, the password should be sent as 0x8745FFFF.

- Reply：OK

Read duress alarm parameter

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x05	0x01		

- **Reply：alarm parameter**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x05	0x01	0x06	parameter

Refer to the writing parameter format.

六、Door timeout alarm parameter

Read

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x06	0x00		

- **Reply：timeout prompt parameter**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x06	0x00	0x04	switch+parameter

Data info：

1byte	2bytes	1byte
Switch	time-out period	Relay output

Switch: 0 means disable, 1 means enable

Time-out period: door sensor opening time. 0 means close, 1-65535 means specified value. Unit is second.

This alarm has no relay output, no records.

Relay output: 0--Non-output relay； 1--output relay(Bandit alarm relay).

Set door timeout prompt parameter

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x06	0x01	0x04	Interface+Switch+parameter

- Reply：OK

七、Door sensor alarm function

Read

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x07	0x00		

- **Reply：alarm parameter**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x7	0x00	0xE1	Parameter

Data info：

1byte	224bytes
Enable	7*8*4 **time zone when the door sensor alarm check is not enabled**

Set door sensor alarm parameter

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x7	0x01	0xE1	Parameter

Reply：OK

八、Legal verification to remove the alarm switch

Setting switch

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x8	0x00	1	Function switch (1)

Reply：OK

Value range：0-- turn off function, 1-- turn on function　default value：1

Read parameter

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x8	0x01	0x00	

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x34	0x4	0x01	1	

九、Remove alarm

Remove alarm

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x04	0x09	0x00	0x02	Alarm types(2)

Alarm Types (multiple choice，1 close alarm，0 do not close alarm):

Value	Explanation
Bit0	Illegal verification alarm
Bit1	Door sensor alarm
Bit2	Duress alarm
Bit3	Door timeout alarm
Bit4	Blacklist alarm
Bit5	Tamper alarm
Bit6	Fire alarm

- Reply：OK

Class 5 Holidays

Preface:

All categories in this class begin with 0x05;

All answers in this class begin with 0x35;

Generally speaking, the Comman and parameter in answer, need the same with Command.

Holiday format:

Part1 (1 byte)	Part2（4 bytes）
Serial No.	Year-Month-Day-Hour

Total 5 bytes

Year: Max. 0-99, 0 means repeat every year.

Hour definition:

Value	Explanation
1	00:00:00 - 11:59:59
2	12:00:00 - 23:59:59
3	00:00:00 - 23:59:59

I. Read holidays capacity info

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x05	0x01	0x00	0x00	

3-Reply: Holiday

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x35	0x01	0x00	0x04	Holiday Information

Info contents:

1st、2nd byte	3rd、4th byte
Holiday capacity	Numbers already stored

二、Clear all holidays

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x05	0x02	0x00	0x00	

Reply：OK

三、Read all holidays

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x05	0x03	0x00	0x00	

4-Reply：Holiday

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x35	0x03	0x00	0x04+0x5*n	Holidays info

Each 5 bytes represents a holiday。

- **Reply：end of transmission**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x35	0x03	0xFF	0x04	Total throughput

四、Add holidays

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x05	0x04	0x00	0x04+5*n	number+serial number

The first 4 bytes represent the number of holidays, and then followed by the holiday info, one holiday per 5 bytes, refer to the beginning of this class for holiday info format.

Reply：OK

五、Delete holiday

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x05	0x05	0x00	0x04+1*n	Quantity+download number

The first 4 bytes represent the number of holidays, and then followed by the holiday info, one holiday per 5 bytes, refer to the beginning of this class for holiday info format.

Reply：OK

Class 6 Opening Time Zone

Preface:

All categories in this class begin with 0x06;

All answers in this class begin with 0x36;

Opening time zone format:

Part1 (2bytes)	Part2 （2bytes）
HH：mm	HH：mm

A group of time zone has 7 days, and a day has 8 time zones, so the number of bytes in a group is：7*8*4=0xE0(224bytes)

The specific definition of the time zone is from left to right, once from Monday to Sunday, each 4 bytes is a small segment, each 32 bytes is a day.

一、Clear all opening time

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x06	0x01	0x00	0x00	

- Reply：OK

二、Read all opening time

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x06	0x02	0x00	0x00	

- **Reply：opening time zone**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x36	0x02	0x00	Group No.(0x2)+0xE0	2+opening time

Transmit all opening time in one time, total is 64 groups, each group is 224 bytes（0xE0）

- **Reply：end of transmission**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x36	0x02	0xFF	0x04	Total throughput

Set as opening time

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x06	0x03	0x00	0xE1	SN(1)+Timezone(224)

- Reply：OK

Class7 Personnel File

Preface:

All categories in this class begin with 0x07;

All answers in this class begin with 0x37;

File format:

<table>
<tr><td>Serial No.</td><td>Byte</td><td>Description</td></tr>
<tr><td>1</td><td>4bytes</td><td>User No.</td></tr>
<tr><td>2</td><td>8bytes</td><td>Card No.
Identification card number 8 bytes，ic、id card 4-3 bytes.</td></tr>
<tr><td>3</td><td>4bytes</td><td>Password
The format is BCD code, i.e. 0x12345678, indicating password is 12345678, password is 8 digits, the value of each digit is 0-9, if the password is less than 8, write the placeholder 0xF in the following space, for example, password is 8765, the password should be sent as 0x8745FFFF</td></tr>
<tr><td>4</td><td>5bytes</td><td>Expiration Date
BCD code, format " Year, Month, Day, Hour, Minute ", Max. 23:59 on December 31, 2099, the second part is automatically 59 seconds.
Example: 0 x9912312359</td></tr>
<tr><td>5</td><td>1byte</td><td>Opening Time
Value Range：1-64，0 means unlimited.</td></tr>
<tr><td>6</td><td>2bytes</td><td>Effective Times
Value range 0-65534， 0 means used up。65535means unlimited</td></tr>
<tr><td>7</td><td>1byte</td><td>User ID<table><tr><td>0</td><td>Common User</td></tr><tr><td>1</td><td>Administrator</td></tr></table></td></tr>
<tr><td>8</td><td>1byte</td><td>Card Type<table><tr><td>0</td><td>Common Card</td></tr><tr><td>1</td><td>Normally Open</td></tr></table></td></tr>
<tr><td>9</td><td>1byte</td><td>Status
0：normal status;
1：Report the loss ；</td></tr>
</table>

		2：Blacklist； 3：Deleted
10	30bytes	User Name
11	30bytes	User No.
12	30bytes	Personnel department
13	30bytes	Personnel Position
14	4bytes	Holiday byte b[3]; b[0].7;Group8 holiday b[0].0;Group1 holiday b[1].7;Group16 holiday b[1].0;Group9 holiday .... b[3].7；Group32 holiday
15	1byte	Enter and exit mark Value： <table><tr><td>Value</td><td>Explain</td></tr><tr><td>3、0</td><td>Effective enter and exit</td></tr><tr><td>1</td><td>Effective enter</td></tr><tr><td>2</td><td>Effective exit</td></tr></table>
16	6bytes	Latest verification time This parameter is set by device， write &HFFFFFFFFFFFF when writing in。 This parameter represents the time when the personnel passed the last verification.
17	1byte	Have fingerprint feature code? 0-Yes；1--No
18	2bytes	Have face feature code? Each bit represents a fingerprint, each person has 10 fingerprints. Bit0--fingerprint 1，bit9--fingerprint 10 0-Yes；1--No

Total 161 bytes 0xA1

一、Read personnel file storage info

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x01	0x00	0x00	

Reply：Authorization card

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x01	0x00	0x18	Authorization card info

info contents：

Capacity info：

Serial No.	Byte	Description
1	4	Maximum capacity of personnel files
2	4	Number of people stored
3	4	Maximum capacity of fingerprint feature code
4	4	Number of fingerprints stored
5	4	Maximum capacity of face feature code
6	4	The number of faces stored

二、Clear all personnel files

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x02	0x00		

Reply：OK

三、Read all personnel files

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x03	0x00		

Reply：personnel file

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data

0x37	0x03	0x00	0x01+0xA1*n	数量+授权卡信息 Quantity + authorization card information

Reply：end of transmission

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x03	0xFF	0x04	总传输量 Total transfer volume

四、Query personnel details

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x03	0x01	0x04	用户号 User ID

此命令读取一条指定ID的人员信息，此命令用于检测某个人的具体权限，用于故障检查。

This command reads a piece of personnel information with a specified ID. This command is used to detect the specific access level of certain person for fault checking.

- 应答：人员详情
- **Reply: Personnel details**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x03	0x01	0xA1	人员档案（参看格式定义）Personnel file(see format definition)

如果没有找到此授权卡那么返回的数据内容为 0xA4 个字节的 0xFF。

If the authorization card is not found, the returned data content is 0xA4 bytes of 0xFF.

Add personnel files

When the user ID already exists, update the information of this user ID

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x07	0x04	0x00	0x01+0xA1*n	Quantity + personnel information*N

- Reply: OK
- **Reply: Failed user ID**

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x37	0x04	0xFF	0x01+0x4*n	Quantity + user ID (4*N)

六、Delete personnel

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x05	0x00	0x01+0x4*n	数量(1B)+用户号（4*N） Quantity (1B) + user ID (4*N)

- Reply: OK

Read personnel files by index

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x13	0x00	5	Serial number 4 bytes+Quantity 1 byte

Serial number is started at 0

- **Reply: personnel file**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x13	0x00	0x01+0xA1*n	Quantity + authorization card information

7. Notify the device to enter the data entry mode

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x20	0x00	6	Type 1 byte + user number 4 bytes + registration number

- **Type: 1. Fingerprint registration; 2. Infrared face registration; 3. Dynamic face registration; 4. Card registration; 5. Password registration;**

Registration serial number: it is only used for fingerprint registration, and the number of registered fingerprints can be set;

Answer: Answer after entering the registration interface (do not wait for the registration to complete)

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x20	0x00	0x01	State

Status: 1. Registration has started; 2. The user number does not exist; 3. The type is wrong or not supported; 4. The serial number is out of range. 5. The storage space of the device is full

- **Enter the registration interface, answer after the operation is completed**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0xF0	0x00	0x01	State

Status: 1. Registration is successful; 2. The user cancels the operation; 3--Repeat registration information

- **When the registered face and the registration in the device are repeated, add a reply indicating the repeated user number**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0xF0	0x01	0x04	Duplicate User ID

九、Personnel Elevator Extended Permissions

Elevator Permission Format

There are 64 groups of elevator permissions, each with 1 byte

port 1 permission	port 2 permission	port 3 permission	....	Port 64 permissions
1 byte	1 byte			

Elevator permission Each byte takes a value of 0 or 1, 0 means no permission, 1 means permission.

Setting permissions

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x07	0x06	0x00	4+64	User number (4 bytes) + 64 bytes permission

- Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x06	0x00	4+1	User number (4) + status

Status description: 1--indicates success; 0--indicates that the user number is not registered; 2--indicates that this function is not supported.

Read permission

Control Code	Data Code

Classification	Command	Parameter	Data length	Data
0x07	0x06	0x01	4	User number (4 bytes)

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x37	0x06	0x01	4+1+64	User number (4) + status + 64 byte authority

Status description: 1--indicates success; 0--indicates that the user number is not registered; 2--indicates that this function is not supported.

Class 8 Records Operation

Preface:

All categories in this class begin with 0x08;

All answers in this class begin with 0x38;

Recording format definition

Recording module code

Value	Explanation
1	Verification records
2	Door sensor records
3	System log
4	Temperature records

Verification recording format

(17 bytes)

4 bytes	4 bytes	6 bytes	1 byte	1 byte	1 byte
Records Serial No.	User No.	Time	Enter/Exit	Status	Photo

Enter/Exit: 1--enter the door; 2--exit the door

Definition of status

Value	Explanation
1	Swiping card verification
2	Fingerprint verification
3	Face verification

4	Fingerprint + Swiping card
5	Face + Fingerprint
6	Face + Swiping card
7	Swiping card + Password
8	Face + Password
9	Fingerprint + Password
10	Manually enter the user number and password for verification
11	Fingerprint+Swiping card+Password
12	Face+Swiping card+Password
13	Face+Fingerprint+Password
14	Face+Fingerprint+Swiping card
15	Repeated verification
16	Expiration date
17	Expiration of opening time zone
18	Can't open the door during holidays
19	Unregistered user
20	Detection lock
21	The effective number has been used up
22	Verify when locked. Do not open the door
23	Reported loss card
24	Blacklist card
25	Open the door without verification -- the user number is 0 when pressing the fingerprint, and the user number is the card number when swiping the card
26	No card swiping verification -- when card swiping is disabled in 【access authentication mode】
27	No fingerprint verification when fingerprint is disabled in [access authentication mode]
28	Controller expired
29	Validation passed - expiration date is coming
30	Body temperature abnormal, refused to enter

Combination mode Record code

Swipe card verification 1
Fingerprint Verification 2
Face Verification 3
Palmprint Verification 36
password authentication 10

****

Swipe + PIN 7

Swipe + Fingerprint 4
Swipe card + face 6
Swipe card + palm print 39

Face + Password 8
Fingerprint + Password 9
Palm print + password 38

Face + palm print 40
Face + Fingerprint 5
****

****
*
Swipe card + face + password 12
Swipe card + fingerprint + password 11
Swipe card + palm print + password 41

Swipe card + fingerprint + face 14
Swipe card + palm print + face 37

Fingerprint + face + password 13
Palmprint + face + password 42

Fingerprint + Palmprint + Face 43
****

Door sensor recording formats

(8 bytes)

First(1b)	Second(6b)	Third(1b)
Door No.	Time	Status

Definition of status:

Value	Explanation
1	Open the door
2	Close the door
3	Enter the door sensor alarm detection status
4	Exit door sensor alarm detection status
5	Door not close completely
6	Use the button to open the door
7	The door is locked when the

	button opens
8	The controller has expired when the button opens the door

System logs recording formats

(8 bytes)

First(1b)	Second(6b)	Third(1b)
Door No.	Time	Status

Definition of status:

Value	Explanation
1	Software unlocked
2	Software locked
3	Software normally open
4	Controller automatically enters into normally open
5	Controller automatically close the door
6	Long press door switch normally open
7	Long press door switch normally close
8	Software locked
9	Software unlocked
10	Controller timing locked--automatically locked on time
11	Controller timing locked--automatically unlocked on time
12	Alarm--Locked
13	Alarm--Unlocked
14	Illegal verification alarm
15	Door sensor alarm
16	Duress alarm
17	Door timeout alarm
18	Blacklist alarm
19	Fire alarm
20	Tamper alarm
21	Remove illegal verification alarm
22	Remove door sensor alarm
23	Remove duress alarm
24	Remove door timeout alarm
25	Remove blacklist alarm
26	Remove fire alarm
27	Remove tamper alarm

28	System is powered
29	System error reset（watchdog）
30	Device formatting records
31	Card reader is connected backwards.
32	Card reader is not connected properly.
33	Unrecognized card reader
34	Network cable is disconnected
35	Network cable has been inserted
36	WIFI is connected
37	WIFI is disconnected

Temperature recording format

(6 bytes)

First(4b)	Second(2b)
Record serial number	Temperature

The temperature is multiplied by 10, 368 is 36.8 degrees, measured in degrees Celsius.

The serial number of the body temperature record and the serial number of the certification record should be the same, that is to say, for every certification record, a body temperature record should be generated.

一、Read the record pointer information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x01	0x00	0x00	

Reply：Info

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x38	0x01	0x00	0x0D*3 supports body temperature device is 0x0D*4 0x0D*4	Records info

info Contents：

Each 0x0D byte is one part，this part is as following：

Part1(4b)	Part2(4b)	Part3(4b)	Part4(1)
Records	Records tail	Upload breakpoint	Cycle mark

capacity	number		

Record the information in the following order:

Part1	Part2	Part3	Part4
Verify records	Door sensor	System record	Temperature Records（Only for temperature products）

二、Clear all records

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x02	0x00	0x00	

- Reply：OK

三、Clear records

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x02	0x01	0x01	Record module code

- Reply：OK

四、Update records pointer

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x03	0x00	0x01+0x04	code+pointer

Pointer format：

Upload the breakpoint
4b

- Reply：OK

五、Update the record tail number

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x03	0x01	0x01+0x04	code+pointer

Pointer Format:

Record the tail number
4b

This Command is primarily used for debugging purposes.

- Reply: OK

六、Read records

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x08	0x04	0x00	0x01+0x04+4	code+serial number+quantity

Data Format:

Records code	Records Serial Number	Read the numbers
1b	4b	4b

- **Reply: Info**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x38	0x04	0x00	0x04+（0x04+0xN）*n	Record info

Records info format:

First (4B)	Second (N)
Records Number	Records Contents

Records Contents:

Part 1.(4b)	Part 2.(nb)
Recording Serial No.	Records contents(Please refer to the records format for details)

- **Reply: end of transmission**

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x38	0x04	0xFF	0x04	Total throuhgput

Class 9 Realtime monitoring info

Preface:

Classes in this class all begin at 0x19;

In addition, all commands in this class, if not marked, are sent by the device

一、Verification Records

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x01	0x00	0x11	verification records

Refer to: verification records format in Class 8 Recording Operation

1.1、Certification record_record with card number

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x01	0x01	0x19	See table below

Record format with card number

4 Bytes	4 Bytes	6Bytes	1Bytes	1Bytes	1Bytes	8Bytes
Record number	User ID	Time	in/out	State	Photo	Card number

二、Door sensor records info

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x02	0x00	0x08	

三、System log Information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x03	0x00	0x08	Records format

Refer to：Body temperature format in Class 8 Records Operation

四、Temperature recording information

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x04	0x00	0x08	Records format

Refer to：Body temperature format in Class 8 Records Operation

五、Connect test package（Pushing Package）

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0x22	0x00	0x00	

This message is on the device as a live packet, when the server parameter in the network parameter is set, the control board will send this message to the server once every second defined by the control board, so that the server can confirm that the control board is online. The SN number is included in the Command [sending information].

六、Network test package

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0xA0	0x00	0x00	

This message is used on the device as a test package to test the state of the network connection, only when the connection test button is pressed

七、Network test package--respond package

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x19	0xA0	0xA0	0x04	Server IP address

This message is sent from the server to the device, and the data packet is immediately fed back to the network test packet when it is received.

Class 10 Remote Upgrade

Note: the maximum firmware value of this protocol is 16M;

I.Prepare to write the upgrade program

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0A	0x01	0x00	4	Firmware file length

Reply：OK

二、Write to the upgrade program file

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0A	0x02	0x00	3+N	Starting position(3)，data(N)

Each instruction sends a firmware data packet block with a maximum length of 256 bytes

2-Reply：Stored successfully

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3A	0x02	0x00		

3-Reply：Not started ready state

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3A	0x02	0x02	0x00	

The firmware file is the encrypted data, each encrypted block is 256 bytes, algorithm RC4, the key is a string: the ASCII code is 5AE27EE392E3E2616CD2F181D7358119

三、After writing to the file, verify CRC32

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0A	0x03	0x00	0x04	The CRC32 value of the firmware

The value of CRC32 is the CRC value of all data after decryption

4-Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3A	0x03	0x00	0x01	1--check successfully 0--check failure

Tips: start the upgrade directly after successful calibration. (First return the check success, and the upgrade will be started immediately after the return)

Class 11 Personnel additional data（Photo， fingerprint， face feature code）and recording photo

I.Ready to write file

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0B	0x01	0x00	6	User No. (4 bytes) File Type (1 byte) Serial No. (1 byte)

File Type：

Value	Command
1	Personnel Photo
2	Fingerprint
3	Infrared face feature code
4	Dynamic face feature code

Serial No.：

The serial number value range of the personnel photo：1-5

The range of fingerprints：0-9

Reply：file handle

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x01	0x00	4	文件句柄 File handle

File handle is used in next writing step, and it can write in file correctly.

When the user number does not exist, the handle returns 0, indicating a reject operation.

二、Write in file

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0B	0x02	0x00	4+3+N	文件句柄（4） 起始位置(3)，数据(N) File handle (4) Start position (3), data (N)

5-Reply：stored successfully

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x02	0x00		

6-Reply：Not started ready state

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x02	0x02	0x00	

三、After writing to the file, verify CRC32

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x03	0x00	0x04	The CRC32 value of the file

7-Reply：

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x03	0x00	0x01	1--check successfully 0--check failure 2--feature code can not be identified 3--personnel photo can not be identified 4--duplicate personnel photo or feature code

Note: validation success and validation failure are only for file CRC32. Validation failure indicates that CRC32 is inconsistent. If the file contents are incorrect, the corresponding code 2 or 3 is returned.

四、Query personnel additional data details

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x04	0x00	0x04	User No.

8-Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x04	0x00	20	User No. (4 bytes) User Photo List (5 bytes) Fingerprint List (10 bytes) Infrared/dynamic face feature code (1 byte)

Return value description:

4 bytes	5 bytes	10 bytes	1 bytes
User No.	Represents the usage situation of 5 photos serial number. The first byte represents the photo with an serial number of 1. The second byte represents the photo with an serial number of 2. And so on Per byte value 1 means Yes, 0 means No	Represents the usage situation of 10 fingerprints serial number. The first byte represents the fingerprint with an serial number of 0. The second byte represents the fingerprint with an serial number of 1. And so on Per byte value 1 means Yes, 0 means No	The storing situation of infrared/dynamic face feature code. 1 means Yes, 0 means No

五、Read personnel photo/recording photo/fingerprint

Read all the contents of the file

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0B	0x05	0x00	6	File type 1 byte Serial No.1 byte (User No./Record the serial number) 4 bytes

File Type:

Value	Command
1	Personnel Photo
2	Fingerprint
3	Record photos
4	Infrared face feature code
5	Dynamic face feature code

Serial No.:

The serial number value range of the personnel photo: 1-5

Fingerprint value range: 0-9

The value of the face feature code is only 1

This parameter is not used when the file type is a recording photo

(User number/record sequence number):

File Type is personnel photo or fingerprint, face feature code, this value represents User No.

File Type is recording photo, and this value represents recording serial number.

Reply: file handle

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x05	0x01	1+4+4+3	File type (1B) Parameters (Card No./Serial No.) (4B) File Handle(4B) File Size(3B)

First return the file handle.

The file handle is 0 if the file does not exist

Reply: file contents

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x05	0x02	4+3+N	File Handle (4) Starting position(3), data(N)

Reply：file contents

The starting position is calculated from 0

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x3B	0x05	0x03	0x04	CRC32校验数据 CRC32 verify data

六、Delete personnel photo/fingerprint/face feature code

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x0B	0x06	0x00	20	User No. (4 Bytes) User photo list (5 Bytes) Fingerprints list (10 Bytes) Infrared/dynamic face feature code (1 Byte)

4 Bytes	5 Bytes	10 Bytes	1 Byte
User No.	Represents the usage of 5 photos serial number. The first byte represents the photo with an serial number of 1. The second byte represents the photo with an serial number of 2. And so on Per byte value 1 means Delete, 0 means Ignore	Represents the usage situation of 10 fingerprints serial number. The first byte represents the fingerprint with an serial number of 0. The second byte represents the fingerprint with an serial number of 1.	The storing situation of infrared/dynamic face feature code. 1 means Delete, 0 means Ignore

Reply：OK

Class 12 Automated test

Automated test instruction

Control Code			Data Code	
Classification	Command	Parameter	Data length	Data
0x80	0x01	0x00	1	Test program code(1)

Program Code	Direction
1	Test screen The screen display sequence: white screen, black screen, red screen, yellow screen, blue screen, rainbow grid image, switch between different images, press any key on the keyboard or tap the screen(touch screen)
2	LED indicator test instruction:the fingerprint access controller quickly switches the green LED and the red LED to flash each other for 1 second (change a color in 0.1 second, and continue for 1 second)
3	Voice test instruction: the fingerprint access controller sends out "Thank you", "Thank you", "Thank you" continuously (3 times)
4	Key test instruction:the screen displays the keys to be tested. The specified key must be pressed three times before it is passed, and then a key is required to be pressed, and all the keys are tested before it is completed. [ESC], [OK],[<-],[->],[1],[2],[3],[4],[5],[6],[7],[8],[9],[0],[*],[MENU] Key sequence: [ESC], [OK],[<-],[->],[1],[2],[3],[4],[5],[6],[7],[8],[9],[0],[*],[MENU]
5	Tamper test instruction: the screen of the fingerprint access controller prompts the tamper test, and wait for the tamper button to be pressed and released three times
6	Card reading test instruction:the screen of the fingerprint access controller shows that please read the card twice, and wait for the card to be read twice (the card number is fixed as 12345678)
7	U disk check test instruction:the fingerprint access controller checks that the U disk reads and writes normally, and automatically imports the personnel information in the U disk
8	Press the fingerprint test instruction: the screen of the fingerprint access controller shows please press the same fingerprint three times, and wait for the same fingerprint to be pressed three times manually (the same user ID is verified three times with the same fingerprint serial number)
9	WG input test instruction: the fingerprint access controller receives the 12345678 card number input by WG, and it is correct twice in a row
10	WG output test instruction: the fingerprint access controller outputs the 12345678 card number through the WG, and outputs twice in a row

11	Touch the screen to detect and display the painted area. It is required to paint the entire screen once to be considered successful
12	WIFI test (automatically search for WIFI information number, even if it can be found successfully)
13	Infrared light test Test whether the infrared light brightness meets the liveness detection standard

Reply : Test completed

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0xB0	0x01	0x00	1	Test result (1)

1--test completed, 0--test failed

Class 13 Remote Upgrade (Dynamic Face Recognition Access Controller)

Note: The maximum value of firmware transmitted by this protocol is 4096M.

I. Ready to write the upgrade program

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0A	0x11	0x00	4	Firmware file length

Reply: OK

二、 Write the upgrade program file

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0A	0x12	0x00	4+N	) Start position (4), data (N)

The maximum length of the firmware data packet block sent by each command is 1024 bytes

Reply: successfully stored

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3A	0x12	0x00		

Reply: not activated ready state

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3A	0x12	0x02	0x00	

The firmware file is encrypted data, each encrypted block is 1024 bytes, the algorithm is RC4, and the key is the string: ASCII code of 5AE27EE392E3E2616CD2F181D7358119

三、After writing the file, perform CRC32 check

Control Code	Data Code

Classification	Command	Parameter	Data Length	Data
0x0A	0x13	0x00	0x04	CRC32 value of the firmware

The value of CRC32 is the CRC value of all data after decryption

Reply:

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3A	0x13	0x00	0x01	1--verification successful 0--verification failed

Tip: Start the upgrade directly after the verification is successful. (First return the verification is successful, and start the upgrade immediately after returning)

Class 14　Read large files

Supports downloading files over 4096M to the computer

一、I. Ready to read files

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0B	0x15	0x00	6	File type 1 byte Serial number 1 byte (User ID/record serial number) 4 bytes

File type:

Value	Command
1	Person avatar
2	Fingerprint
3	Record photos
4	Infrared face feature code
5	Dynamic face feature code

Serial number:

The value range of the serial number of the person's avatar: 1-5

Fingerprint value range: 0-9

The value range of the facial feature code is only 1

This parameter is not used when the file type is recording photos

(User ID/record serial number):

When the file type is: person's avatar or fingerprint, face feature code, this value represents the user number

The file type is: when recording photos, this value represents the record serial number.

Reply: File handle

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3B	0x15	0x01	1+4+4+4+4	File type (1B) Parameters (user ID/serial number) (4B)

				File handle (4B) File size (4B) File CRC32 (4B)

First return the file handle.

File handle: the context associated number for subsequent file reading. The handle is released after receiving [Stop 0x0B1504].

If the file does not exist, the file handle is 0

二、Read file block

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0B	0x15	0x02	4+4+2	File handle (4) Start position (4), Number of reads (2)

The maximum read size of each block is 4096 bytes,

The starting position is calculated from 0.

Reply: File content

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3B	0x15	0x02	4+4+2+4+N	file handle (4) starting position(4) Data block length(2) CRC32 check data (4) File content (N: 0-4096)

Return all block contents at once, and the data length can be up to 4200.

三、Release the read file handle

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0B	0x15	0x03	4	File handle (4)

After the read file handle is released, you can no longer use this handle to read the contents of the file block

Reply: The file handle has been closed

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x3B	0x15	0x03	0	

The sixteenth class roll call function

1、Notify the device to enter the roll call state

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0D	0x01	0x00	2	The duration of entering the roll call state

Duration: 0-65535 seconds, 0 means waiting until someone is recognized successfully, then exit the roll call state
Other means, wait for the specified number of seconds, and automatically exit the roll call state after the time is up
Reply:OK

After entering the roll call state, wait for the personnel to perform face recognition. After the recognition is successful, send a monitoring message of successful recognition [Class Ninth Monitoring Message] - [1. Authentication Record], the record code is 3 -- face authentication
When the identification timeout, send【Ninth monitoring message】-【3. System log】, the record code is 39 -- roll call timeout

2、Notify device to go to sleep

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0D	0x02	0x00	0	

Notify the device to go to sleep immediately
Reply：OK

3、Notify the device to play the specified voice

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0x0D	0x03	0x00	1	voice index number

Attachment 1 Voice

Voice Content	Description	Broadcast Way
Welcome, come in	When the identity is entering the door	Only once
Thank you, bye!	When the identity is to go out	Only once
Verification failed, please try again	Remind when verification fails	Only once
Please shut door	Overtime Remind	Repeat every 30 seconds
Fire, please leave quickly	Repeat every 10s	Repeat every 10 seconds
Access level expired	Remind if user access level expired	Only once
You cannot open the door at this time	Remind if the user does not have the access level for this period	Only once
System locked	Valid card remind when system locked	Only once
Connection error	Remind if RS485 or WEIGEND Error	Only once
The user has been reported as lost	Remind during authentication	Only once
Password error, please re-enter	Remind if admin or card password error	Only once
Door normal open	Remind if door in normal open status	Only once
Soft-Pulse open successfully	Remind if Soft-Pulse Open	Only once
Formatting now	Remind before format hardware	Only once
Reading interval is too small	Reading time too short remind after setting reading interval	Once
Enter password	Remind if the card need to enter password	Only once
The user has been registered as a "blacklist"	Remind if user is in the blacklist during authentication	Only once
Access level expired soon	Expire time remind (30 days in advance)	Only once

Attachment 2 Priority of the alarm type

1.Fire Alarm
2.Force Alarm
3.llegal Authentication Alarm
4.Door Sensor Alarm
5.Blacklist Alarm
6.Overtime Alarm
7.Tamper Alarm
For example: After any alarm is activated, the fire alarm is activated again, and the fire alarm is the current status.

Because the priority of the fire alarm is too high, other alarms cannot be activated when the fire alarm has been activated.

Attachment 3 UDP search

When searching for UDP, use UDP broadcast to search for devices.
The device only receives: [Search different network identification devices], [Set device network identification], [Read network parameters], [Write network parameters]
In UDP mode, the target SN in the received command is not judged, and the source SN needs to be changed to the SN of the device in the returned command, and the communication password is fixed 0xFFFFFFFF.

Attachment 4 Device initialized value

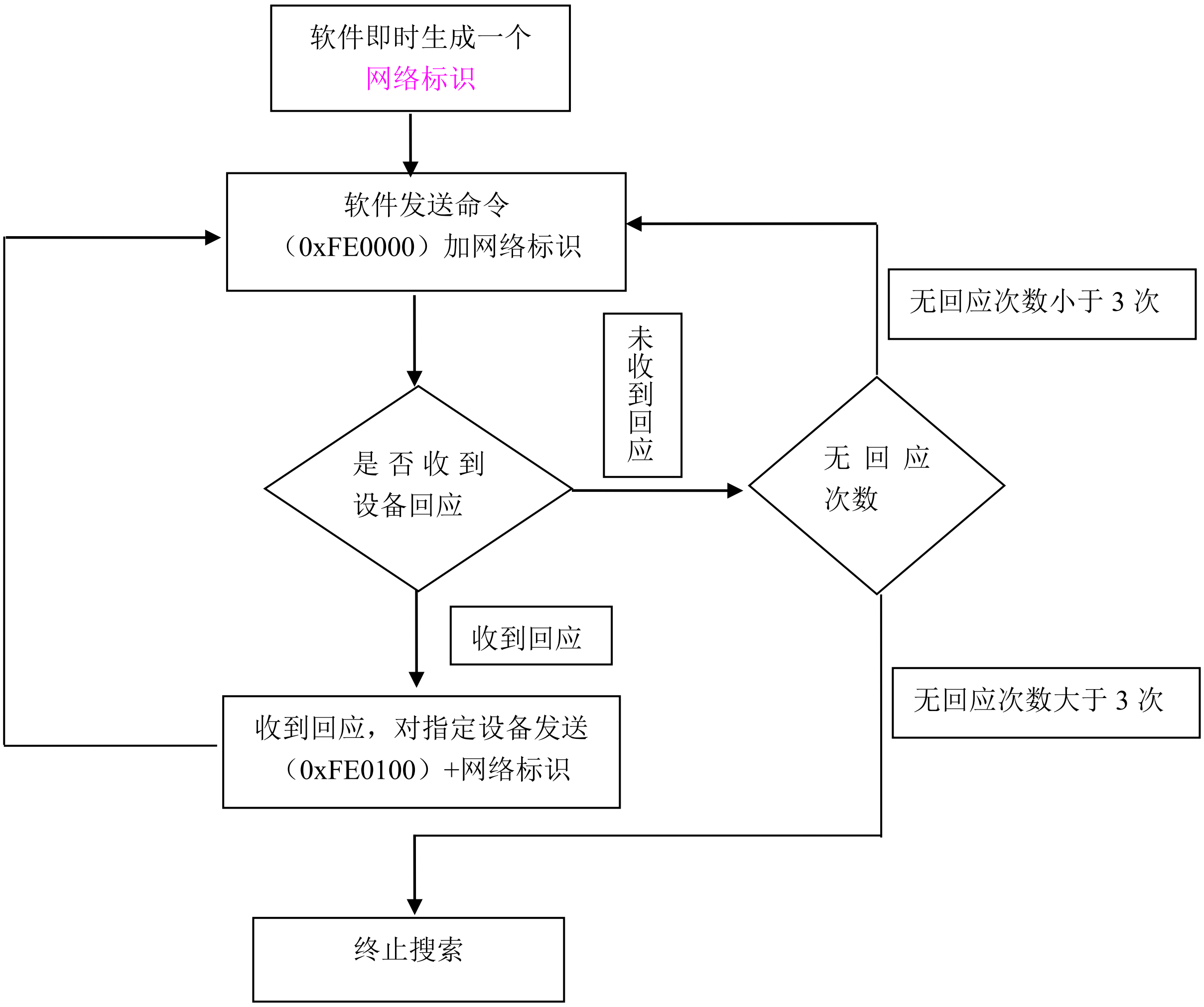
- Set the storage mode of the memory: automatically cycle storage after the storage is full

B formatting does not change the data part: (the data cannot be moved when formatted)

- SN address code cannot be changed
- Communication password cannot be changed
- The number of days the hardware is valid cannot be changed
- All information of the IP address cannot be changed
- The number of operating days of the system cannot be changed
- The number of formatting cannot be changed
- The number of watchdog resets cannot be changed

Attachment 5 Automatically search for devices

Automatic search steps:



Attachment 6 Encrypted communication

After the encrypted communication is enabled, the controller switches to be the encrypted communication mode. In this mode, only read SN and search SN commands are executed in non-encrypted mode, and other commands need to enter a special decoding process.

Note: After the encrypted communication is enabled, the fixed-line communication password of the data packet is 0xFFFFFFFF, and this password is used for commands that do not need to be encrypted.

In order to be compatible with the old version for the protocol, the encrypted communication command puts the encrypted data in the data part of the old protocol. The control code is cured as

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
0xFC	0x98	0x9A	X+2	Encrypted data + verification code (2)

Control Code			Data Code	
Classification	Command	Parameter	Data Length	Data
1Byte	1Byte	1Byte	4Byte	Variable length

The verification code uses the CRC16 algorithm and occupies 2 bytes.

The CRC16 value in the verification code indicates the CRC16 value before data encryption. After the receiver is required to decrypt the data, the decrypted data should be CRC checked. Correctly indicates that the decryption is successful. This verification code is used to verify whether the data is correct after decryption. Incorrect reply to the password error in the predefined command (this reply is not encrypted).

When the controller reply to commands, if it only needs to reply to predefined commands (success, password error, verification error), encryption processing is not required.

The communication encryption switch is affected by the initialization command, and the initialization communication encryption switch will be released, that is, the communication is not encrypted.

In the encrypted communication process, the communication password segment required in the original communication instruction is not used, and 0xFF is used instead. The real communication password does not appear in the communication instruction.

Encrypted instructions include message-type instructions such as swiping card monitoring from the controller. [Connection Confirmation] and [Connection Test] in the message command do not need to be encrypted.

Obtaining SN instructions, reading network parameters, searching for devices, setting network flags, these instructions are not controlled by the encryption switch.

See Attachment《RC4 Encrypted Communication Command Design Scheme.txt》

Correspondence Table of Holidays in Personnel File

Binary
Send holiday data
Actual order of use
No.32
NO.1
Convert to hexadecimal

编码说明
Unicode Coding description
a1 ： 0x4E2D00610031
Middle a1: 0x4E2D00610031

Appendix 7 Weather Icon Correspondence Table

daytime weather	night weather	Simplified	daytime weather	night weather	Simplified
0	30	Sunny	5	5	Ice particles
0	30	Sunny	6	6	Sleet
0	30	Sunny	6	6	Sleet
0	30	Sunny	7	7	light rain
0	30	Sunny	7	7	light rain
0	30	Mostly sunny	8	8	Moderate rain
0	30	Mostly sunny	9	9	Heavy rain
1	31	Partly cloudy	10	10	Rainstorm
1	31	Partly cloudy	10	10	Heavy rain
1	31	Partly cloudy	10	10	Torrential rain
1	31	Partly cloudy	14	14	light snow
1	31	Few clouds	14	14	light snow
2	2	Negative	15	15	Moderate snow
2	2	Negative	15	15	Moderate snow
3	33	Shower	16	16	Heavy snow
3	33	Shower	17	17	Blizzard

3	33	Shower	19	19	Freezing rain
3	33	Shower	19	19	Freezing rain
3	33	Shower	7	7	light rain
3	33	Partial shower	8	8	moderate rain
3	33	Partial shower	9	9	Heavy rain
3	33	light shower	10	10	Heavy rain
3	33	Strong shower	10	10	Heavy rain
13	34	Snow showers	14	14	light snow
13	34	Small snow showers	14	14	light snow
18	32	Fog	14	14	light snow
18	32	Fog	15	15	Heavy snow
18	32	Freezing fog	15	15	Heavy snow
20	36	Sandstorm	16	16	Heavy snow
29	35	Dust	15	15	Snow
29	35	Dust devil	8	8	Rain
29	35	Jansa	45	46	Haze
20	36	Strong sandstorm	1	31	Partly cloudy
45	46	Haze	1	31	Partly cloudy
45	46	Haze	1	31	Partly cloudy
2	2	Negative	18	32	Fog
4	4	Thunderstorm	18	32	Fog
4	4	Thunderstorm	2	2	Negative
4	4	Thunderstorm	3	33	Shower
4	4	Thunderstorm	4	4	Thunderstorm
4	4	Thunderstorm	4	4	Thunderstorm
4	4	lightning	4	4	Thunderstorm
4	4	lightning	4	4	Thunderstorm
5	5	Thunderstorm with hail	7	7	light to moderate rain
5	5	Thunderstorm with hail	9	9	Moderate to heavy rain
5	5	Hail	10	10	Torrential rain

5	5	Ice needle	15	15	light to moderate snow

Appendix 8 Door Opening QR Code

QR code door opening data structure:

Field	Data length
Card number	9 Bytes
Expiration date	4 Bytes (compressed time format)(second 6b, minute 6b, hour 5b, day 5b, month 4b, year 6b)
Check	1Byets CRC8 **Data verification before encryption**

QR code Due to the characteristics of the QR code, the data needs to be encrypted with RC4, and the key is a string:

The ASCII code of 1e30b3ec0f634874956e27627dfc2c46

The data of the QR code is the value of these 14 bytes.

The card reader sends fifteen bytes, the first one is 0x2E, which means the QR code. This is followed by 13 bytes of real data of the QR code and a checksum of 1 byte.

The starting year of the year is 2018; that is, the value of the year is added to the value of 2018 to equal the value of the final year.

Appendix 9 Guest Password
Password calculation method: "month day * 6663 + year month day + visitor root password, then take the last 10 as the first 8 digits)
For example: October 10 * 6663 + October 10, 2022 * 88888888 (the root password set by the customer) = 2,395,612,420,488,320 (the password to open the door is equal to 2,420,488,3)
If the month and day are single digits, no need to add 0

October 13, 2022 = 1013*6663+20221013+99999999= 126,970,631 (take the first 8 digits = 12697063)

December 31, 2022 = 1231*6663+20221231+99999999= 128,423,383 (take the first 8 digits = 12842338)

January 1, 2099 = 11*6663+209911+99999999= 100,283,203 (take the first 8 digits = 10028320)

Appendix 10 IC Card Sector Dynamic Password
The algorithm idea of dynamic password is:
1. First generate a newkey (10 bytes) based on the confusion of the sector password (6 bytes) and the card number (4 bytes)

2. Then confuse the newkey to generate a 6-byte newdata.
3. Use rc4 encryption, the key is newkey, the plaintext is newdata, and the real sector password is the encrypted ciphertext.

The source code is as follows:

```
/* use RC4 encryption after obfuscating the card number and key
return value:
1 -- Indicates that the encryption is complete
-1 -- Indicates that the key length is less than 6 bytes
-2 -- Indicates that the length of the cardcode is less than 4 bytes
-3 -- Indicates result_len length is less than 6 bytes
*/
RC4_API long WINAPI Create_Section_Password(BYTE *key, long key_len, BYTE *cardcode, long cardcode_len, BYTE *SectionPWD, long SectionPWD_len)
{
if (key_len < 6)
{
return -1;
}
if (cardcode_len < 4)
{
return -2;
}
if (SectionPWD_len < 6)
{
return -3;
}

BYTE newKey[10];
BYTE newData[6];
newKey[6] = key[0];
newKey[4] = key[1];
newKey[8] = key[2];
newKey[5] = key[3];
newKey[2] = key[4];
newKey[0] = key[5];

newKey[1] = cardcode[0];
newKey[7] = cardcode[1];
newKey[3] = cardcode[2];
newKey[9] = cardcode[3];

newData[0] = newKey[1];
newData[1] = newKey[9];
newData[2] = newKey[5];
newData[3] = newKey[6];
```

```
newData[4] = newKey[7];
newData[5] = newKey[3];

RC4_Encrypt(newKey, 10, newData, 6, SectionPWD, 6);
return 1;}
```

Appendix 11 IC Card Data Structure--User Card

Serial number	Content	Data length	
1	Password	4 Bytes	0-3
2	In and out identification; 0----in, 1----out, other---in and out	1 Bytes	4
3	Validity period; BCD code, year month day	3Bytes	5,6,7
4	Door opening period; 1-64	1Bytes	8
5	Valid door opening times; 0-65535; 0--invalid; 65535--unlimited	2Bytes	9,10
6	Each permission card type Bit0, bit1 indicate permission 1 Bit2, bit3 indicate permission 2 Bit4, bit5 indicate permission 3	1Bytes	11
7	Authority 1 (3-byte area address + 9-byte gate address table)	12Bytes	
8	Authority 2 (3-byte area address + 9-byte gate address table)	12Bytes	
9	Authority 3 (3-byte area address + 9-byte gate address table)	12Bytes	

Permission format:

1	Level 1 identification code, valid when the card type is level 2, 3, or 4	1Bytes
2	Level 2 identification code, valid when the card type is level 3 or level 4	1Bytes
3	Level 3 identification code, valid when the card type is level 4 authority	1Bytes
4	**Address Data Sheet** **The data in this table has different identification codes corresponding to different permission levels.** **Each bit represents an address, and a total of 72 addresses can be represented.** **The order of address bits represents 1-72 addresses from high to low.** **The highest bit of the first byte indicates address 1. The lowest bit of the last byte indicates address 72.**	9Bytes

Card type 2bit. Value meaning: 0--Level 4 authority, 1--Level 3 authority, 2--Level 2 authority, 3--Level 1 authority

Level 4 permissions:

Indicates that 4 fields need to be verified: Level 1 identification code, Level 2 identification code, Level 3 identification code, address data (level 4 identification code)

When swiping the card to open the door, the controller needs to verify whether the 1st, 2nd, and 3rd-level identification codes match the machine, and then verify whether the bit corresponding to the 4th-level identification code in the address data is 1 after matching.

Level 3 permissions:

Three fields need to be verified: Level 1 identification code, Level 2 identification code, address data (level 3 identification code)

When swiping the card to open the door, the controller needs to verify whether the 1st and 2nd-level identification codes match the machine, and then verify whether the bit corresponding to the 3rd-level identification code in the address data is 1 after matching. (The level 3 identification code is used in the address table for level 3 authority.) This type of authority does not verify the level 4 identification code

Level 2 permissions:

Two fields need to be verified: level 1 identification code, address data (level 2 identification code)

When swiping the card to open the door, the controller needs to verify whether the level 1 identification code matches the machine, and then verify whether the bit corresponding to the level 2 identification code in the address data is 1 after matching. (The level 2 identification code is used in the address table for level 2 authority.) This type of authority does not verify the level 3 and level 4 identification codes

Level 1 permissions:

1 field needs to be verified: address data (level 1 identification code)

When swiping the card to open the door, the controller only verifies whether the bit corresponding to the first-level identification code in the address data is 1. (A level 1 identification code is used in the address table for level 1 authority.) This type of authority does not verify level 2, 3, and 4 identification codes

Appendix 12 IC Card Data Structure--Parameter Setting Card

Set the card data format (the first sector starts with the first block):

	setting type（1B）	set data length（2B）	data segment（nB）	Data check 1 byte
Access control parameter setting card	0XD0	88	88B parameter data	Sum check
Card number permission setting card	0XD1	7+4*n(0<n<175)	1B card type + 5B card validity period + 1B door opening period + N card data	Sum check
Door opening time setting card	0XD2	n	N time slots	Sum check

Sum check: type, length, data segment, all bytes are added to remove the low byte.

Access control parameter setting card

Communication SN code	8 Bytes		0-7
communication password	4Bytes		8-11
manage password	5Bytes	1 byte switch, 4 byte password	12-16
Door opening delay time	2Bytes	After the door is opened, how long is the delay to turn off the relay. 0 means -0.5 seconds, the maximum is 65535 seconds	17-18
Door open timeout parameter	4Bytes	1 byte switch, 2 byte time, 1 byte alarm output switch	19-22
Alarm time	1Bytes	0XFF----always alarm, 0 no alarm.	23
Alaceholder	1Bytes		24
Communication switch	1Bytes	0---do not communicate with software, 1---communicate with software	25
Monitor switch	1Bytes	0---do not open real-time monitoring, 1---open real-time monitoring	26
Illegal card alarm switch	1Byte	0---Normal, 1---Read an illegal card and send an alarm	27
Whether the record is full or not	1Byte	0---cycle, 1---not cycle	28
Card reading interval	2Bytes	Switch (1) + Spacer Card Quantity (1)	29-30
placeholder	2 Bytes		31-32
Time Calibration Parameters	2Bytes		33-34
Blacklist alarm switch	1Byte	0---no alarm, 1---alarm	35
IC card parameters	19Bytes		36-54
IC setting card parameters	8Bytes		55-62
Added on April 25, 2017		Add 25 bytes	
Voice broadcast switch	10Bytes		63-72
Offline B-type door opening card	7Bytes	Function switch 1 byte + sector password 6 bytes	73-79
Prohibition of using IC card switch	1Byte	Disables use of Philips S50 and S70 cards when opened 1 or 0	80
485communication encryption switch	1Byte	After opening, the communication uses RC4 encryption, refer to the protocol document 1 or 0	81
IC card dynamic password switch	1Byte	After opening, the IC card sector password is dynamically calculated according to the card number 1 or 0	82
Clear switch after parameter setting card is used	1Byte	After opening, the card needs to be cleared after the card is imported once.	83
Type B card with card number will be cleared after import	1Byte	After opening, import the card number and clear the card number flag in the card.	84
Relay output type	1Byte	1--COM & NC, 2--COM & NO, 3--Bistability	85
Exit switch parameters	2Bytes	Enable switch 1 byte, allow normally open 1 byte	86-87

There are 88 byte parameters in total.

Card settings.

card type	1Byte	0--ordinary door opening card; 1--report lost card; 2--blacklist card	0
Expiry dateBCD	5Bytes	year month day time	1-5
Opening hours	1Byte	1 byte switch, 4 byte password	6
Authorization card number	4*NBytes		7-N

Each card has 4 bytes.

Card number (4B)
Card number (4B)

Each setting card can set up to 175 cards.

Opening hours

The format of each opening period is changed to:

Time slot number (1B)	Monday session length	Content on Monday	....	Sunday session length	Sunday content
Indicate which paragraphs need to be set	（1B）	N Byte		（1B）	N Byte

The content of the time slot is the front part of the time slot, and the content of the time slot every day of the week must not exceed 8*4 = 32 Byte. If it is less than 32 Byte, it will be filled with FF.

The advantage of this setting is that if the customer only uses 1-2 time slots and each time slot has only 2-3 time slices per day, 64 time slots can be set at a time (62 time slots are all FF)

Type Time Package

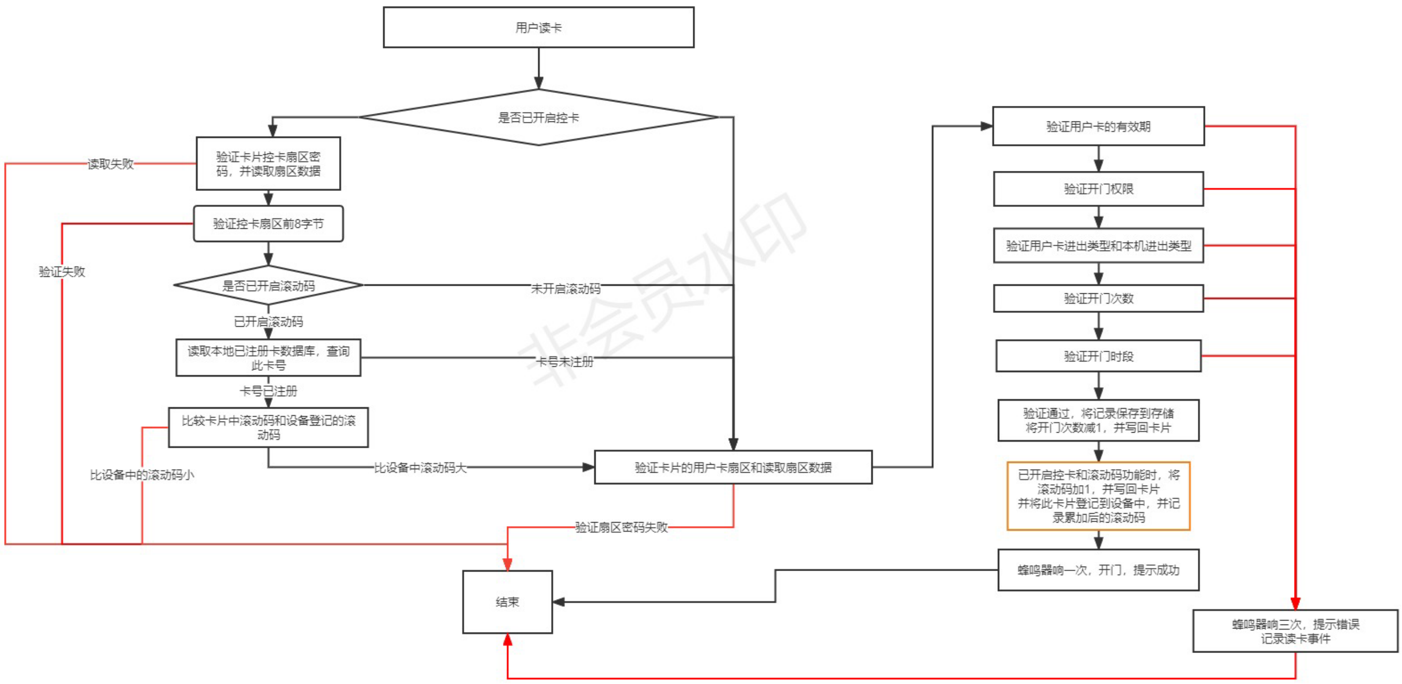
D2 Number of time slots Each time slot

Time period Monday Tuesday

Number of time periods Content of time periods Number of time periods Content of time periods.....

Note: In the period package, the data length is not set 附 Record 13 IC card rolling code function

1. The rolling code function can only be enabled after the control card function is turned on
2. The rolling function starts from the first byte in the first block of the control card sector, and takes 4 bytes as the rolling code, indicating the number of times the card has been swiped in this system
3. After verifying the card control parameters of the card, the device reads the rolling code and compares it with the list of registered cards locally. If the serial number of the last card read is smaller than the locally registered serial number, If it means that the card may have been copied, ignore this card reading


用户读卡
是否已开启控卡
验证卡片控卡扇区密码，并读取扇区数据
读取失败
验证控卡扇区前8字节
验证失败
是否已开启滚动码
未开启滚动码
已开启滚动码
读取本地已注册卡数据库，查询此卡号
卡号未注册
卡号已注册
比较卡片中滚动码和设备登记的滚动码
比设备中的滚动码小
比设备中滚动码大
验证卡片的用户卡扇区和读取扇区数据
验证扇区密码失败
结束
验证用户卡的有效期
验证开门权限
验证用户卡进出类型和本机进出类型
验证开门次数
验证开门时段
验证通过，将记录保存到存储
将开门次数减1，并写回卡片
已开启控卡和滚动码功能时，将滚动码加1，并写回卡片
并将此卡片登记到设备中，并记录累加后的滚动码
蜂鸣器响一次，开门，提示成功
蜂鸣器响三次，提示错误
记录读卡事件